

VMware NSX-T

Interview Questions & Answers

300 Professional Q&A | 10 Topic Sections

300

Questions

10

Sections

2026

Edition

Created by Akhilesh Chaudhari

Senior IT Engineer | MCA | TechShiksha

akhileshchaudhari.com

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Table of Contents

1. NSX-T Architecture & Core Concepts	Q 1–53 (53)
2. Logical Switching & Overlay Networking	Q 54–86 (33)
3. Logical Routing — Tier-0 & Tier-1 Gateways	Q 87–118 (32)
4. NSX Edge Nodes & Edge Clusters	Q 119–146 (28)
5. NSX-T Distributed Firewall (DFW)	Q 147–175 (29)
6. NSX-T Load Balancer	Q 176–197 (22)
7. NSX-T VPN — IPsec & L2 VPN	Q 198–218 (21)
8. NSX-T Security — Advanced Features	Q 219–243 (25)
9. NSX-T Operations & Troubleshooting	Q 244–270 (27)
10. NSX-T Multi-Cloud & Advanced Topics	Q 271–300 (30)

NSX-T Architecture & Core Concepts

53 Q&A;

Q 1

What is VMware NSX-T Data Center?

Answer

VMware NSX-T Data Center is a software-defined networking (SDN) and security platform that virtualizes the entire networking stack — logical switches, routers, load balancers, firewalls, VPNs — independent of the underlying hardware. Unlike NSX-V (which was tied to vSphere), NSX-T supports multiple hypervisors (ESXi, KVM), bare-metal servers, public clouds, and container platforms (Kubernetes). It enables micro-segmentation, zero-trust security, and consistent networking policy across any workload.

Q 2

What is the difference between NSX-T and NSX-V?

Answer

NSX-V (NSX for vSphere) was tightly integrated with vSphere and vCenter — it could only manage VMware ESXi hosts. NSX-T (NSX Transformers) is hypervisor-agnostic — it supports ESXi, KVM, bare metal, and containers. NSX-T has its own management plane (NSX Manager) independent of vCenter, uses Geneve encapsulation (instead of VXLAN in NSX-V), and supports multi-cloud environments. NSX-V is End-of-Life; NSX-T is the current and future platform.

Q 3

What are the three planes in NSX-T architecture?

Answer

NSX-T uses a three-plane architecture: (1) Management Plane — NSX Manager cluster provides the UI, API, and configuration storage; it is the single source of truth for all NSX-T configuration. (2) Control Plane — Central Control Plane (CCP) manages the distribution of networking state (routes, MAC/IP tables, firewall rules) to data plane components via the Local Control Plane (LCP) on each transport node. (3) Data Plane — the actual packet forwarding happens here on transport nodes (ESXi hosts, KVM, Edge nodes) using the NSX kernel modules (n-VDS / VDS).

Q 4

What is NSX Manager and what does it do?

Answer

NSX Manager is the centralized management appliance for NSX-T. It provides the HTML5 UI, REST API, and policy framework for configuring all NSX-T components. NSX Manager runs as a cluster of 3 appliances (for HA) or a single node (for lab). It manages the full lifecycle of logical switches, routers, firewalls, load balancers, VPNs, and security groups. It also integrates with vCenter (for vSphere environments), Kubernetes (for container networking), and public cloud connectors.

Q 5

What is NSX Manager cluster and why is 3-node recommended?

Answer

NSX Manager is deployed as a cluster of 3 virtual appliances for high availability and scale. The 3-node cluster uses a Raft consensus algorithm – all three nodes participate in leader election. If the active manager fails, one of the remaining nodes automatically becomes the leader without any manual intervention. A 3-node cluster can tolerate one node failure and maintain full functionality. A single-node manager is supported only for proof-of-concept or lab environments and cannot tolerate any failure.

Q 6

What is a Transport Node in NSX-T?

Answer

A Transport Node is any host or device that participates in the NSX-T data plane – it can send and receive overlay (Geneve-encapsulated) traffic. Transport nodes include: ESXi hosts (Host Transport Nodes), KVM hosts, bare-metal servers, and NSX Edge nodes. Each transport node runs the NSX kernel module (NSX vSwitch / n-VDS) and has at least one TEP (Tunnel Endpoint) IP address used for overlay tunneling between nodes.

Q 7

What is a TEP (Tunnel Endpoint) in NSX-T?

Answer

A TEP (Tunnel Endpoint) is the IP address on a transport node used to create Geneve tunnels to other transport nodes. When a VM on Host A sends traffic to a VM on Host B, the NSX kernel module on Host A encapsulates the packet in a Geneve header using Host A's TEP IP as source and Host B's TEP IP as destination. The physical network routes the Geneve-encapsulated packet between TEPs. TEPs are assigned dedicated VMkernel ports (on ESXi) on a dedicated TEP VLAN/subnet.

Q 8

What is Geneve encapsulation and how does it differ from VXLAN?

Answer

Geneve (Generic Network Virtualization Encapsulation) is the overlay encapsulation protocol used in NSX-T. It is more flexible than VXLAN – Geneve supports variable-length option fields in its header, allowing metadata to be carried alongside the encapsulated packet (used by NSX-T for security context, service chaining metadata). VXLAN (used in NSX-V) has a fixed header with only a 24-bit VNI. Both use UDP (Geneve: port 6081, VXLAN: port 4789). Geneve is the IETF standard for future-proof overlay networking.

Q 9

What is an N-VDS (NSX Virtual Distributed Switch)?

Answer

N-VDS is the NSX-T native virtual switch that runs on transport nodes (ESXi/KVM hosts). It is responsible for the NSX data plane on the host — handling overlay (Geneve) encapsulation/decapsulation, distributed routing, and distributed firewall enforcement. In vSphere 7.0+, N-VDS was merged with the vSphere Distributed Switch (VDS 7.0) — so a single VDS can handle both vSphere and NSX-T traffic. In older environments, a separate N-VDS coexists alongside the vDS.

Q 10

What is the NSX-T Policy API vs Manager API?

Answer

NSX-T has two API frameworks: Policy API (preferred for new deployments) — uses a declarative, intent-based model where you define desired state and NSX-T figures out how to implement it; objects are organized hierarchically under Infra/Projects; simplifies configuration with automatic realization. Manager API (legacy) — uses an imperative model where you specify exact configuration steps; more granular control but more complex. Policy API is recommended for all new NSX-T 3.0+ deployments. Some advanced features still require Manager API.

Q 11

What are Segments in NSX-T?

Answer

Segments (called Logical Switches in older NSX-T versions) are the NSX-T equivalent of VLANs or port groups — they provide Layer 2 broadcast domains for VMs and workloads. Segments can be: Overlay Segments (use Geneve encapsulation — span multiple hosts across any IP network, not VLAN-limited), VLAN Segments (carry a specific VLAN tag — used for uplink connections to physical networks, or for workloads that need native VLAN access). VMs connect to segments via logical ports.

Q 12

What is a Segment Profile in NSX-T?

Answer

Segment Profiles define port-level settings applied to logical ports on a segment. Types: QoS Profile (bandwidth limits, DSCP marking), Spoofguard Profile (prevents IP/MAC spoofing on ports), MAC Discovery Profile (controls MAC learning and limit), IP Discovery Profile (controls IP address discovery via ARP/NDP/DHCP snooping), and Security Profile (used with port-level security policies). Profiles are reusable — attach the same profile to multiple segments to ensure consistent policy.

Q 13

What is the Overlay Transport Zone in NSX-T?

Answer

A Transport Zone defines the scope of overlay networking – which transport nodes can participate in the same overlay fabric. An Overlay Transport Zone includes hosts that can exchange Geneve-encapsulated traffic. Any segment created in an overlay transport zone automatically spans all hosts in that transport zone. Transport zones are attached to transport node profiles and applied to hosts during host preparation. A transport node can belong to multiple transport zones (overlay and VLAN).

Q 14

What is a VLAN Transport Zone?

Answer

A VLAN Transport Zone is used for segments that carry VLAN-tagged traffic instead of overlay (Geneve) traffic. VLAN segments in this zone are used for: NSX Edge uplinks to physical routers (connecting Tier-0 to the physical network), VLAN-backed workloads that cannot use overlay networking, and management traffic. VLAN transport zones are typically attached to Edge nodes and uplink profiles that define the physical NIC used for VLAN connectivity.

Q 15

What is Uplink Profile in NSX-T?

Answer

An Uplink Profile (called Host Switch Profile in Manager API) defines the physical NIC binding, teaming policy, MTU, and transport VLAN for a transport node's N-VDS/VDS. Key settings: Teaming Policy (active-active or active-standby with named uplinks like uplink-1, uplink-2), MTU (must be at least 1700 for Geneve overlay – recommended 9000 with jumbo frames), and Transport VLAN (the VLAN ID for TEP traffic on the physical network). Uplink profiles are reusable across multiple transport nodes for consistent configuration.

Q 16

What is the recommended MTU for NSX-T overlay networks?

Answer

The recommended MTU for NSX-T overlay networks is 1600 bytes minimum (to accommodate the Geneve overhead of ~50 bytes on top of a 1500-byte inner packet), but VMware strongly recommends using jumbo frames with MTU 9000 end-to-end (from transport nodes through all physical switches to Edge nodes). Using 9000 MTU eliminates fragmentation, improves performance for large data transfers (vMotion, storage, application data), and is a requirement for NSX-T production deployments. Physical switches must also support the configured MTU.

Q 17

What is BFD in NSX-T and why is it used?

Answer

BFD (Bidirectional Forwarding Detection) is a lightweight protocol used by NSX-T to detect failures in the data plane path between transport nodes and Edge nodes. NSX-T uses BFD over the TEP tunnel between hosts and Edges to quickly detect tunnel failures (typically sub-second). When BFD detects a failure, it triggers route convergence faster than waiting for routing protocol timers. BFD is enabled by default on Tier-0 and Tier-1 routers for fast failover between Edge cluster members.

Q 18

What is IPFIX in NSX-T?

Answer

IPFIX (IP Flow Information Export) is a flow monitoring protocol supported by NSX-T that exports IP flow records (source/destination IP, port, protocol, byte/packet counts) from NSX logical switches and distributed firewalls to an external IPFIX collector. It provides visibility into East-West traffic inside the overlay – traffic that never exits to the physical network and is invisible to traditional network monitoring tools. IPFIX in NSX-T supports per-segment and per-port monitoring with configurable sampling rates.

Q 19

What are the deployment models for NSX Manager?

Answer

NSX Manager supports three deployment sizes: Small (2 vCPU, 8 GB RAM – lab only), Medium (4 vCPU, 16 GB RAM – production up to 1024 transport nodes), Large (12 vCPU, 48 GB RAM – production up to 10,000+ transport nodes). For production, always deploy a 3-node cluster (one on each ESXi host in the management cluster) for HA. The cluster VIP (Virtual IP) is used for API/UI access and automatically fails over between nodes if the active manager fails.

Q 20

What is NSX Intelligence?

Answer

NSX Intelligence (now part of NSX Advanced Threat Prevention) is an analytics engine that provides visibility, recommendation, and micro-segmentation planning for NSX-T environments. It continuously collects flow data from NSX-T transport nodes and builds an application traffic map showing all communication flows between VMs. Security administrators can use NSX Intelligence to: discover actual application communication patterns, automatically generate recommended firewall rules based on observed traffic, identify anomalous traffic, and plan micro-segmentation policies before enforcing them.

Q 21

What is the NSX-T REST API and how is it used?

Answer

NSX-T provides a comprehensive REST API (both Policy API and Manager API) for automating all configuration and operational tasks. The API uses JSON payloads over HTTPS. Authentication: Basic Auth (username/password) or Principal Identity (certificate-based for automation). Common use cases: automated provisioning of segments/routers/firewall rules via Terraform (VMware NSX-T provider), Ansible (community.vmware.nsx collection), or custom scripts; integration with ITSM tools (ServiceNow); CI/CD pipeline automation for network changes. The API Explorer at <https://nsx-manager/policy/api/v1/spec> provides interactive documentation.

Q 22

What is vSphere with NSX-T integration?

Answer

In vSphere environments, NSX-T integrates with vCenter to enable VM-level networking and security. During host preparation, NSX-T installs its kernel module (VIBs) on ESXi hosts and creates the N-VDS or uses VDS 7.0+. VMs connect to NSX-T segments via logical ports. NSX-T reads VM inventory from vCenter (VM names, tags, security groups) to apply dynamic security policies. Integration is configured in NSX Manager under System > Fabric > Compute Managers by adding vCenter as a compute manager.

Q 23

What is a Principal Identity in NSX-T?

Answer

A Principal Identity (PI) is a certificate-based identity used for machine-to-machine authentication with the NSX-T API. Instead of embedding username/password credentials in automation scripts, you create a PI with a client certificate, and automation tools authenticate using that certificate. PIs are used by: VMware vRA/Aria Automation, Kubernetes NCP (NSX Container Plugin), Terraform, and other orchestration tools. PIs support role assignments (full access, read-only, network operator) and can be scoped to specific NSX-T projects.

Q 24

What is NSX Federation?

Answer

NSX Federation (introduced in NSX-T 3.0) provides multi-site network and security management from a single Global Manager appliance. The Global Manager synchronizes configuration (stretched segments, gateway firewall policies, security groups) across multiple Local Manager instances (one per site). This enables: consistent security policy across data centers, stretched Layer 2 segments between sites (for workload mobility), global security groups spanning multiple sites, and centralized visibility. Federation is used in active-active and active-standby multi-site deployments.

Q 25

What is the NSX Container Plugin (NCP)?

Answer

NCP (NSX Container Plugin) is the CNI (Container Network Interface) plugin that integrates NSX-T with Kubernetes. NCP runs as a pod in each Kubernetes cluster and uses the NSX-T Policy API to: automatically create NSX-T logical ports for each pod, provide pod networking via overlay segments, implement Kubernetes Network Policies as NSX distributed firewall rules, and integrate NSX-T load balancer with Kubernetes Services (ClusterIP, NodePort, LoadBalancer type). NCP enables consistent NSX-T security policies to apply to both VMs and containers.

Q 26

What are the hardware requirements for NSX Manager?

Answer

NSX Manager appliance requirements: Small (lab): 2 vCPU, 8 GB RAM, 200 GB disk. Medium (production): 4 vCPU, 16 GB RAM, 200 GB disk. Large (production scale): 12 vCPU, 48 GB RAM, 200 GB disk. All sizes require a dedicated management network with a static IP and FQDN. For a 3-node cluster: all three managers should be on different ESXi hosts (DRS anti-affinity rule recommended) and ideally on different datastores for storage failure resilience. The cluster VIP must be on the same subnet as the individual manager IPs.

Q 27

What is NSX-T used for in a modern data center?

Answer

NSX-T provides complete software-defined networking for modern data centers: Network Virtualization (logical switches, routers, load balancers replacing physical hardware), Micro-segmentation (granular firewall rules per VM/pod to limit lateral movement), Multi-cloud Networking (consistent policies across on-premises, VMware Cloud on AWS, Azure VMware Solution), Container Networking (Kubernetes pod networking via NCP/Antrea), SD-WAN (branch connectivity), Zero-Trust Security (distributed firewall with identity-based policies), and Network Automation (REST API, Terraform, Ansible integration).

Q 28

What is Antrea and its relation to NSX-T?

Answer

Antrea is an open-source Kubernetes CNI plugin developed by VMware that uses Open vSwitch (OVS) for pod networking and implements Kubernetes NetworkPolicy. NSX-T provides the Antrea-NSX integration — Antrea runs as the CNI in Kubernetes clusters and reports flow/policy data to NSX-T for centralized security visibility and policy management. In vSphere with Tanzu (Supervisor Cluster), Antrea is the default CNI and integrates with NSX-T for distributed firewall enforcement on pod traffic.

Q 29

What is the difference between NSX-T 3.x and NSX 4.x?

Answer

NSX 4.x (rebranded from NSX-T Data Center) introduced several enhancements: Project Northstar multi-tenancy framework, enhanced Federation capabilities, NSX+ cloud-delivered management (SaaS), improved vSphere 8.0 integration with VDS 8.0, enhanced AI/ML-based threat detection (NSX Advanced Threat Prevention), simplified installation with new NSX Installer, and improved distributed firewall with identity-based tagging. NSX 4.x retains full backward compatibility with NSX-T 3.x configurations. The product is now simply called 'VMware NSX' (dropping the -T suffix).

Q 30

What is the NSX-T VIP (Virtual IP) and how does it work?

Answer

The NSX Manager cluster VIP is a floating IP address shared by the 3-node manager cluster. All UI and API access goes through the VIP. The active manager node holds the VIP and responds to requests. If the active manager fails or is shut down, the remaining nodes elect a new leader within seconds, and the VIP automatically moves to the new leader's interface. No load balancer is needed for manager HA – the VIP provides seamless failover. The VIP must be on the same L2 subnet as the individual manager IP addresses.

Q 31

What is NSX-T tag and its use in security policies?

Answer

NSX-T Tags are key-value metadata labels (e.g., Tag: 'Scope=Environment, Value=Production') that can be applied to VMs, segments, groups, and other objects. Tags are the foundation of dynamic security groups – a group can be defined as 'all VMs with Tag Scope=App, Value=WebTier'. When a VM is tagged, it is automatically included in matching groups and subject to all firewall policies applied to those groups. Tags can be applied manually (UI/API) or automatically via vCenter custom attributes, Active Directory group membership, or VM name patterns.

Q 32

What is the NSX-T CCP (Central Control Plane)?

Answer

The Central Control Plane (CCP) is a component embedded in NSX Manager that maintains the network state and distributes it to transport nodes. The CCP receives topology changes (VM migrations, new segments, route updates) and pushes updated MAC/IP tables, routing tables, and firewall rules to the Local Control Plane (LCP) on each transport node. The CCP uses a messaging bus (NSX RPC/MQ) to communicate with LCPs. If CCP is unreachable (NSX Manager down), existing transport nodes continue forwarding traffic using cached state – but no new configuration changes are applied.

Q 33

What is LCP (Local Control Plane) in NSX-T?

Answer

The Local Control Plane (LCP) runs on each transport node (ESXi host kernel module / Edge node) and maintains the local forwarding state. The LCP receives updates from the CCP (MAC tables, routing tables, firewall rules) and programs the data plane (N-VDS/VDS) accordingly. The LCP also reports local state changes (VM power-on/off, MAC/IP discovery) back to the CCP. If the CCP connection is lost, the LCP continues operating with cached state – existing traffic continues to flow uninterrupted, but new configuration changes are not applied until connectivity to CCP is restored.

Q 34

What ports does NSX-T Manager use?

Answer

NSX Manager key ports: TCP 443 (HTTPS – UI and REST API), TCP 9443 (Manager appliance management – VAMI), TCP 5671 (RabbitMQ/AMQP – messaging bus between manager and transport nodes), TCP 1234 and 1235 (NSX Manager cluster inter-node communication), UDP 4789 (VXLAN – if used), UDP 6081 (Geneve – overlay traffic between TEPs), TCP 179 (BGP on Edge nodes), TCP 500 / UDP 4500 (IKE/IPsec VPN on Edge), TCP 22 (SSH – management). Physical firewalls between NSX components must allow these ports for proper NSX-T operation.

Q 35

What is the NSX-T Certificate Management?

Answer

NSX-T uses TLS certificates for: NSX Manager UI/API (server certificate presented to browsers/clients), inter-node communication (cluster certificates), and communication with transport nodes (NSX Manager signs certificates for transport nodes). Certificates are managed in NSX Manager: System > Certificates. You can: replace the NSX Manager self-signed certificate with a CA-signed certificate for production, renew expiring certificates (NSX Manager warns 30 days before expiry), and manage principal identity certificates for automation. Expired NSX Manager certificates cause UI/API access failures – plan certificate renewal before expiry.

Q 36

What is the Transport Node Profile?

Answer

A Transport Node Profile is a template in NSX-T that defines the configuration applied to host transport nodes: which N-VDS/VDS to use, uplink profile, TEP IP pool, and transport zone membership. Instead of configuring each ESXi host individually, you create a Transport Node Profile and apply it to a vSphere Cluster — NSX-T automatically prepares all current and future hosts in that cluster as transport nodes using the profile settings. This enables automated, consistent host preparation and is especially useful when new ESXi hosts are added to a cluster (they are automatically onboarded as NSX-T transport nodes).

Q 37

What is the NSX-T compute manager?

Answer

A Compute Manager is an external system (typically vCenter Server) registered with NSX Manager to provide compute inventory (hosts, clusters, VMs, datacenters). NSX-T uses the compute manager to: discover ESXi hosts for transport node preparation, read VM attributes (tags, names) for DFW dynamic groups, apply Transport Node Profiles to vSphere clusters, and track VM migrations (vMotion) for firewall rule portability. A single NSX Manager can have multiple compute managers (multiple vCenter servers). Register via: NSX Manager > System > Fabric > Compute Managers > Add.

Q 38

What is NSX-T HA (High Availability) for NSX Manager?

Answer

NSX Manager HA is achieved through a 3-node cluster. All three nodes are active and share the configuration database (replicated via Raft consensus). The cluster has a Virtual IP (VIP) used for all management access — the active leader node holds the VIP. If the leader node fails, the remaining two nodes elect a new leader within seconds, and the VIP moves automatically. There is no manual failover required. The cluster can tolerate one node failure and continue fully operational. Two-node failure causes the cluster to become read-only (split-brain protection). For production, always deploy 3 NSX Manager nodes on different ESXi hosts and datastores.

What is the NSX-T Upgrade Coordinator?

The NSX-T Upgrade Coordinator is built into NSX Manager and guides administrators through the NSX-T upgrade process. It provides: pre-upgrade checks (compatibility, configuration validation), component upgrade sequencing (manager first, then Edge nodes, then hosts), progress tracking for each component, pause/resume capability (upgrade one Edge or host cluster at a time), rollback option (if upgrade fails mid-process). The Upgrade Coordinator ensures the correct upgrade order is followed – upgrading hosts before NSX Manager or Edges can cause compatibility issues. NSX-T upgrade bundles (.mub files) are uploaded to NSX Manager before starting the coordinator.

What is NSX-T Node VM Deployment?

NSX-T component VMs (NSX Manager, Edge nodes) are deployed as OVA/OVF appliances. NSX Manager can be deployed using the NSX Manager installer GUI (browser-based), CLI installer, or Terraform/PowerCLI automation. Edge nodes are deployed from NSX Manager UI (System > Fabric > Nodes > Edge Transport Nodes > Add) or via API. For automated deployments, VMware provides the NSX-T Terraform Provider (registry.terraform.io/providers/vmware/nsxt) and Ansible collection ([community.vmware.nsxt](https://community.vmware.com/guides/nsx-t-ansible)) to deploy and configure NSX-T components as Infrastructure as Code.

What is NSX-T configuration backup frequency recommendation?

VMware recommends NSX-T configuration backups: minimum daily scheduled backup to a remote SFTP server. Additionally: take a manual backup before any major change (upgrade, large topology change, security policy update). Retain at least 3 backup versions (current + 2 previous) to allow rollback options. Verify backup integrity regularly by checking backup timestamps in NSX Manager. Store backups on a different server than NSX Manager – if NSX Manager's hardware fails, you need the backup on a separate system for restore. Include NSX-T backup verification in your DR runbook and test restore periodically.

Q 42

What is NSX-T Global Manager vs Local Manager in Federation?

Answer

In NSX Federation: Global Manager (GM) is a separate NSX Manager cluster deployed for cross-site management. Administrators configure global objects (stretched segments, global firewall policies, global groups) on the GM, which synchronizes them to Local Managers. Local Manager (LM) handles site-specific resources and is the data plane controller for its site. Each production site has its own LM. The GM provides a unified view across all sites. If GM is unavailable, LMs continue operating independently – federation is a management-plane feature, not a data-plane dependency.

Q 43

What is NSX-T Edge Cluster load balancer distribution?

Answer

When multiple T1 gateways with LB services are connected to the same Edge cluster, NSX-T distributes the LB Service Router (SR) instances across Edge nodes for load balancing. Each T1's SR is assigned to one Edge node (based on the T1's placement during creation). To achieve even distribution: create T1 gateways in round-robin order (each new T1 is typically placed on the least-loaded Edge node). For dedicated LB workloads, consider creating a separate Edge cluster for LB T1 gateways – isolating LB traffic from routing traffic. Monitor Edge node CPU and memory to identify imbalanced LB distribution and manually re-balance by migrating SR instances.

Q 44

What is NSX-T Encrypted Traffic Inspection?

Answer

NSX-T Advanced Threat Prevention supports TLS/SSL encrypted traffic inspection via the Distributed Firewall's SSL Proxy feature. SSL Proxy decrypts TLS traffic at the hypervisor level (at the VM's vNIC) for inspection by IDS/IPS signatures and L7 App-ID, then re-encrypts it before forwarding. This is different from SSL offloading at the LB – SSL Proxy is inline in the DFW datapath. Configuration requires importing the CA certificate used to re-sign decrypted connections. Privacy considerations: SSL decryption should only be performed where legally permitted and with proper data handling policies. Selective decryption (only specific traffic categories) is recommended to balance security and performance.

Q 45

What is NSX-T installation pre-requisites checklist?

Answer

Before installing NSX-T: verify vSphere version compatibility (vSphere 6.7U3+ for NSX-T 3.x, vSphere 7.0+ recommended), confirm ESXi CPU supports hardware virtualization (VT-x/AMD-V), ensure management network has static IPs with DNS and NTP configured, verify MTU 9000 on physical switches for TEP traffic, confirm TEP VLAN is trunked on all host uplink switch ports, validate firewall rules allow NSX Manager ports (443, 5671), plan IP addressing (Management IPs for 3 NSX Managers, TEP IP pool, Edge uplink IPs), size NSX Manager VMs correctly (at least Medium for production), and download NSX-T OVA from VMware Customer Connect.

Q 46

What is the NSX-T API versioning?

Answer

NSX-T REST API versions: Policy API v1 (current, at `/policy/api/v1/`) is the primary API for all new configurations. Manager API v1 (legacy, at `/api/v1/`) is available for backward compatibility and advanced operations. API versioning means specific endpoints may be added or deprecated between NSX-T versions. Always check the NSX-T API Guide for the specific version in use. API backward compatibility is maintained within major versions (3.x APIs work on all 3.x releases). When upgrading NSX-T, verify any custom automation scripts against the new version API guide for deprecated endpoints. Use `/api/v1/spec` and `/policy/api/v1/spec` for interactive API exploration.

Q 47

What is NSX-T password policy?

Answer

NSX-T password policy for local accounts: minimum length (default 12 characters), complexity requirements (uppercase, lowercase, digits, special characters), password expiry (configurable, default 90 days), password history (cannot reuse last 5 passwords), account lockout (default 5 failed attempts, 15-minute lockout). Configure via NSX Manager: System > User Management > Password Policy. For production security: increase minimum length to 16+, enable lockout, and integrate with AD via vIDM to use enterprise password policies instead of local policies. Service accounts (Principal Identities) use certificates — no password policy applies to them.

Q 48

What are common NSX-T deployment mistakes to avoid?

Answer

Common NSX-T deployment mistakes: (1) MTU not set to 9000 end-to-end – causes intermittent large-packet drops in overlay. (2) No DRS anti-affinity for Edge VMs – Edge nodes on same host fail together. (3) TEP and management on same VMkernel – noisy neighbor issue, MTU mismatch risk. (4) Using Policy API and Manager API for same objects – creates inconsistency, support issues. (5) Default Allow DFW policy in production – no zero-trust enforcement. (6) No VCSA or NSX Manager backup configured – no recovery path if appliance fails. (7) Edge VMs on non-HCL hardware – performance and support issues. (8) Insufficient Edge VM size for LB workload – Small/Medium Edge cannot handle large LB environments. (9) Not planning route redistribution – T1 subnets not reachable from physical network.

Q 49

What is NSX-T sizing guidance for large environments?

Answer

NSX-T sizing for large environments (1000+ VMs): (1) NSX Manager: deploy Large appliances (12 vCPU, 48 GB RAM) in 3-node cluster for up to 10,000+ transport nodes. (2) Edge Nodes: Large Edge (8 vCPU, 32 GB) for high routing throughput; XL Edge (16 vCPU, 64 GB) for large LB deployments (200K+ concurrent connections). (3) DFW: monitor ESXi host CPU – DFW adds ~5-10% per vNIC; budget 20-30% ESXi CPU headroom for DFW overhead at scale. (4) TEP IP Pool: /24 minimum per cluster (254 usable IPs). (5) BGP routes: T0 can handle 1M+ routes in RIB. (6) Firewall rules: keep total DFW rules under 50K for performance headroom below the 100K maximum.

Q 50

What is NSX-T API idempotency?

Answer

NSX-T Policy API is idempotent – sending the same PUT/PATCH request multiple times produces the same result as sending it once. This is critical for automation: if a Terraform apply or Ansible playbook is re-run (e.g., after a partial failure), it will not create duplicate objects or cause errors – it will simply confirm the desired state is already in place and make no changes. Idempotency is achieved via path-based object identification (/infra/segments/my-segment) – the path uniquely identifies the object, so re-creating the same path overwrites/updates the existing object rather than creating a duplicate. Manager API is not fully idempotent – use Policy API for automation.

Q 51

What is NSX-T NTP configuration requirement?

Answer

Accurate NTP synchronization is critical for NSX-T: certificate validity checks require correct time, log timestamps must be accurate for SIEM correlation and forensics, and cluster consensus (Raft algorithm in NSX Manager) requires all nodes to have synchronized time. Configure NTP: on NSX Manager (System > Settings > NTP), on Edge nodes (set `ntp-server` command in Edge CLI), and ensure ESXi hosts (hosting NSX Manager and Edge VMs) also use the same NTP servers. All NSX-T components should sync from the same NTP source hierarchy. Maximum allowed time drift between NSX components is typically <5 seconds — larger drift can cause certificate validation failures and cluster instability.

Q 52

What is NSX-T product versioning and support lifecycle?

Answer

NSX-T product lifecycle: NSX-T 2.x (EOL), 3.0/3.1/3.2/3.3 (LTS — Long Term Support, 5 years support), NSX 4.x (current, renamed from NSX-T). VMware follows an N-1 support policy for most versions (current + previous major). General Support: 5 years from GA. Technical Guidance: additional 2 years (limited support). Always run a supported version — check End of General Support dates on vmware.com/support/lifecycle. For production environments: use LTS versions for stability, upgrade within 12 months of a new major release for security patches, and plan upgrade projects 6 months before current version reaches End of General Support.

Q 53

What is NSX-T deployment on KVM?

Answer

NSX-T supports KVM (Kernel-based Virtual Machine) hypervisors as transport nodes alongside ESXi hosts. KVM host preparation: install the NSX-T VIBs (Debian/RPM packages) on the KVM host manually or via NSX Manager automation. The N-VDS on KVM uses Open vSwitch (OVS) as the underlying switch. KVM transport nodes support overlay segments, distributed routing (DR), and DFW — the same features as ESXi transport nodes. Use cases: environments with mixed ESXi and KVM hosts (e.g., OpenStack deployments on KVM with some vSphere workloads), or organizations standardizing on NSX-T for consistent networking across heterogeneous hypervisor environments.

Logical Switching & Overlay Networking

33 Q&A;

Q 54

How does NSX-T Overlay networking work end-to-end?

Answer

When VM-A on Host-1 sends a packet to VM-B on Host-2: (1) VM-A sends a standard Ethernet frame. (2) The NSX kernel module on Host-1 looks up the destination MAC in its ARP/MAC table (populated by the control plane). (3) The packet is encapsulated in a Geneve header with Host-1's TEP IP as source and Host-2's TEP IP as destination. (4) The Geneve packet is sent over the physical IP network (UDP port 6081). (5) Host-2's NSX kernel module receives the Geneve packet, decapsulates it, and delivers the inner Ethernet frame to VM-B. The physical network only sees IP packets between TEPs.

Q 55

What is ARP suppression in NSX-T?

Answer

ARP suppression prevents unnecessary ARP broadcast flooding across the overlay network. In traditional networks, when VM-A needs VM-B's MAC address, it broadcasts an ARP request to all hosts in the broadcast domain. In NSX-T, the distributed dataplane learns IP-to-MAC mappings and stores them centrally. When VM-A sends an ARP request, the local NSX kernel module on the host responds directly with VM-B's MAC (proxy ARP) without flooding the request across all transport nodes. This dramatically reduces broadcast traffic in large overlay networks.

Q 56

What is VTEP and how does it differ from TEP?

Answer

In older NSX documentation, VTEP (VXLAN Tunnel Endpoint) referred to the tunnel endpoint used with VXLAN encapsulation (NSX-V). In NSX-T, the equivalent is called TEP (Tunnel Endpoint) and uses Geneve encapsulation. A TEP is an IP address assigned to a VMkernel port (on ESXi) or a dedicated NIC (on KVM/bare metal) used exclusively for overlay tunnel traffic. Functionally they serve the same purpose — originating and terminating overlay tunnels — but TEP/Geneve is more flexible than VTEP/VXLAN and is the current standard in NSX-T.

Q 57

What is a Logical Port in NSX-T?

Answer

A Logical Port is a virtual port on an NSX-T Segment that a VM or workload connects to. Each logical port has its own security profile, QoS policy, and statistics. Logical ports can be in different states: Up (connected and active), Down (disconnected), or Blocked (blocked by security policy). In NSX-T Policy API, logical ports are automatically created when a VM's vNIC is connected to a segment. Logical ports track the attachment point (which host a VM is running on) and notify the control plane so routing/firewall tables are updated.

Q 58

What is a VNI (Virtual Network Identifier) in NSX-T?

Answer

VNI is the 24-bit identifier in the Geneve header that identifies the logical network (segment) to which the encapsulated traffic belongs. Each NSX-T Segment gets a unique VNI (similar to a VLAN ID, but 24-bit – allowing over 16 million unique segments vs 4094 VLANs). The NSX Manager allocates VNIs from a configurable pool. When a Geneve packet arrives at a transport node, the VNI tells the NSX kernel module which logical segment the inner frame belongs to, so it can be delivered to the correct VM or forwarded correctly.

Q 59

What is the difference between Overlay and VLAN segments?

Answer

Overlay Segments use Geneve encapsulation – traffic between VMs is tunneled over any IP network, independent of physical VLANs. Overlay segments can span multiple hosts across different racks/data centers. They are used for VM-to-VM communication within the NSX fabric. VLAN Segments are backed by a specific physical VLAN – they are used for connecting NSX Edge nodes to the physical network (uplink segments) or for workloads that need to be on a specific physical VLAN (e.g., management VMs that must communicate with physical devices). VLAN segments require trunk-configured physical switch ports.

Q 60

What is MAC learning in NSX-T segments?

Answer

MAC learning in NSX-T segments allows the logical switch (segment) to dynamically learn the MAC addresses of connected workloads – similar to how a physical switch learns MACs. In NSX-T, MAC learning is configured via the MAC Discovery Profile attached to a segment. It controls: whether MAC learning is enabled (needed for nested virtualization, containers, or NIC bonding inside VMs), the MAC learning table limit, and whether unknown unicast traffic is flooded or dropped. For most VM workloads, MAC learning is disabled because the control plane already knows MAC-to-TEP mappings.

Q 61

What happens when a VM is vMotioned in NSX-T?

Answer

When a VM is live-migrated (vMotion) from Host-A to Host-B in NSX-T: (1) NSX-T detects the VM migration via the vCenter API. (2) The logical port for the VM is moved from Host-A to Host-B in the NSX control plane. (3) The TEP mapping for that VM's MAC/IP is updated on all relevant transport nodes. (4) The distributed firewall rules follow the VM to Host-B automatically. (5) Any existing TCP connections are maintained because the VM's IP/MAC do not change. The entire process is transparent to the VM and its connections.

Q 62

What is the RTEP (Remote Tunnel Endpoint) in NSX Federation?

Answer

RTEP (Remote Tunnel Endpoint) is used in NSX Federation for inter-site traffic. While regular TEPs handle intra-site overlay traffic, RTEPs are dedicated endpoints on Edge nodes used for creating Geneve tunnels between two NSX-T sites. RTEPs allow stretched segments (Layer 2 extension across sites) and inter-site routing in a federated NSX deployment. RTEPs are configured on dedicated Edge interfaces with dedicated IP addresses on the inter-site network (typically over an MPLS or internet WAN link between data centers).

Q 63

What is the maximum number of segments supported in NSX-T?

Answer

NSX-T supports up to 100,000 logical ports per NSX Manager cluster and up to 10,000 segments per transport zone (as of NSX-T 3.x/NSX 4.x). The 24-bit VNI space allows over 16 million unique segment IDs theoretically, but practical limits are defined by the NSX Manager's database capacity and the configuration maximums in the VMware documentation. The NSX-T Configuration Maximums guide (available at configmax.esp.vmware.com) should always be consulted for specific version limits before designing large deployments.

Q 64

What is the TEP Pool and how is it configured?

Answer

The TEP Pool (IP Pool) is a pre-configured range of IP addresses allocated to transport nodes for their TEP interfaces. When a host is prepared as a transport node, NSX-T automatically assigns an IP from the TEP pool to the host's VMkernel TEP port. TEP pools are configured in NSX Manager under Networking > IP Management > IP Pools. The TEP pool subnet should be on a dedicated VLAN (TEP VLAN) that is routable between all transport nodes. Configuring the TEP pool on a dedicated network ensures overlay traffic is isolated from management and VM traffic.

Q 65

What is an NVDS profile and how is it different from an uplink profile?

Answer

An N-VDS profile (Host Switch Profile in Manager API, or Uplink Profile in Policy API) defines the configuration applied to the N-VDS on a transport node, including: physical NIC (uplink) binding names (uplink-1, uplink-2), active/standby teaming policy per traffic type, transport VLAN for TEP traffic, and MTU. In Policy API, the Uplink Profile is the equivalent object. When preparing a host as a transport node, you attach an uplink profile that tells NSX-T which physical NICs to use and how to configure them for NSX traffic.

Q 66

What is EVPN and does NSX-T use it?

Answer

EVPN (Ethernet VPN) is a BGP-based control plane for distributing MAC/IP reachability information in overlay networks. NSX-T does not use EVPN internally – it uses its own CCP (Central Control Plane) to distribute MAC/IP and routing information to transport nodes. However, NSX-T Tier-0 gateways can peer with EVPN-enabled physical switches/routers using BGP EVPN for data center gateway connectivity (route-type 5 IP prefix advertisement). This allows NSX-T overlay routes to be redistributed into the physical EVPN fabric seamlessly.

Q 67

What is the BUM traffic handling in NSX-T overlay?

Answer

BUM (Broadcast, Unknown-unicast, Multicast) traffic handling in NSX-T overlay: Broadcast – ARP suppression handles most ARP broadcasts; remaining broadcast is replicated to all segment members via the control plane-directed replication. Unknown unicast – NSX-T uses ingress replication (head-end replication) by default – the source host replicates the packet to all other TEPs in the segment (no multicast required on physical network). Multicast – NSX-T can use ingress replication (no physical multicast required) or hybrid/multicast replication modes where physical multicast is available. Ingress replication is the default and most common deployment model.

Q 68

What is segment discovery profile?

Answer

A Segment Discovery Profile (IP Discovery Profile) controls how NSX-T learns the IP addresses of workloads connected to a segment. Methods: ARP Snooping (learns IP from ARP packets), DHCP Snooping (learns IP from DHCP ACK packets), VMware Tools (reads IP from VM guest info via vCenter – most reliable for vSphere VMs), and NDP Snooping (for IPv6). Accurate IP discovery is critical for Distributed Firewall rules that reference IP addresses, for Spoofguard enforcement, and for NSX Intelligence flow analysis. Combining DHCP Snooping + VMTools gives the most accurate IP discovery.

Q 69

What is Spoofguard in NSX-T?

Answer

Spoofguard is an NSX-T security feature that prevents IP and MAC spoofing on logical ports. It maintains a whitelist of valid IP/MAC addresses per logical port (learned via IP Discovery). When a VM sends a packet with a source IP or MAC not in its whitelist, Spoofguard drops the packet. Spoofguard is configured via a Spoofguard Profile attached to a segment. It has two modes: Port Bindings (enforces the learned/approved IP-MAC pairs) and Whitelist (allows any IP for the port — effectively disabled). Spoofguard is critical for preventing VM-level network attacks.

Q 70

What is the difference between N-VDS and VDS in NSX-T?

Answer

N-VDS (NSX Virtual Distributed Switch) is NSX-T's native virtual switch, deployed independently of vSphere VDS. Pre-vSphere 7.0, NSX-T created a separate N-VDS alongside the existing vDS on ESXi hosts — requiring NICs to be split between vDS and N-VDS. From vSphere 7.0+, the VDS was enhanced to support NSX-T natively (VDS 7.0 = N-VDS convergence). A single VDS handles both vSphere and NSX-T traffic, eliminating the need for a separate N-VDS. This simplifies NIC management and is the recommended approach for new vSphere 7.0/8.0 + NSX-T deployments.

Q 71

What is Cross-vCenter NSX networking?

Answer

Cross-vCenter NSX (now called NSX Federation) allows NSX segments and policies to span multiple vCenter Servers. In traditional NSX-V, this required a Universal Logical Switch with a Universal Transport Zone. In NSX-T, cross-vCenter networking is achieved via NSX Federation — a Global Manager synchronizes configurations across Local Managers (one per vCenter). Stretched segments, global security groups, and unified firewall policies can span all sites managed by the Global Manager. This enables consistent networking and security for multi-data center or multi-vCenter environments.

Q 72

How does NSX-T handle unknown unicast traffic?

Answer

NSX-T handles unknown unicast (traffic to a destination MAC not in the local MAC table) using head-end replication (ingress replication) by default. The source transport node replicates the unknown-destination packet to all other TEPs in the segment. Each receiving transport node checks if the destination VM is local; if not, it discards the packet. This avoids requiring physical multicast support on the underlay network. For large segments with many hosts, the replication overhead can be significant. NSX-T's ARP suppression minimizes unknown unicast by proactively populating MAC/IP tables.

Q 73

What is a Segment Port Profile in NSX-T?

Answer

A Segment Port Profile aggregates multiple profile types (QoS, IP Discovery, MAC Discovery, Security, Spoofguard) into a single reusable object that can be attached to a segment or a logical port. Instead of configuring each profile type individually on every segment, create a Segment Port Profile combining all relevant profiles and apply it once. This simplifies management in large environments with many segments requiring the same port-level settings. Segment Port Profiles are created in NSX Manager > Networking > Segments > Segment Profiles and can be applied at the segment level (default for all ports) or overridden at the individual port level.

Q 74

What is NSX-T segment bridge (VLAN bridging)?

Answer

NSX-T Segment Bridging provides Layer 2 connectivity between an NSX-T overlay segment and a VLAN on the physical network. This allows VMs on an NSX-T overlay segment to communicate at Layer 2 with physical servers or VMs on a VLAN-backed network (without routing). Bridging is implemented on Edge nodes via a Bridge Profile. Use cases: migrating workloads from VLAN to overlay gradually (brownfield), providing L2 access to legacy physical systems that cannot be reached via routing. Bridging should be a transitional solution — long-term architecture should use routing (T0/T1 gateways) rather than L2 bridging for inter-network communication.

Q 75

What is the NSX-T Default Gateway for a segment?

Answer

In NSX-T, the default gateway for VMs on a segment is the T1 gateway's interface IP on that segment (configured when you connect the segment to the T1). The T1 creates a logical interface (downlink) on the segment with the gateway IP address (e.g., 10.0.1.1/24 for a 10.0.1.0/24 segment). This IP is configured in VM guest OS as the default gateway. The T1's distributed router (DR) component on each ESXi host responds to ARP requests for the gateway IP (proxy ARP) so VMs can resolve it locally without sending ARP requests off-host. This is part of NSX-T's distributed routing.

Q 76

What are NSX-T Segment MAC address limits?

Answer

Each NSX-T segment port has a configurable MAC address learning limit to prevent MAC table flooding attacks (a VM inside the overlay generating thousands of fake MAC addresses to fill the MAC table). The limit is configured in the MAC Discovery Profile attached to the segment. Default limit: 4096 MACs per port. For segments with nested hypervisors (running VMs inside VMs) or containers (where many MAC addresses appear on one port), increase this limit. When the limit is exceeded, new MAC addresses are blocked (traffic from unknown MACs is dropped) and an alarm is generated in NSX Manager.

Q 77

What is NSX-T Multicast support?

Answer

NSX-T overlay networking does not natively support IP multicast between VMs in the overlay (unlike physical networks). Multicast traffic from VMs is typically dropped at the overlay boundary. For workloads requiring multicast: (1) Use VLAN-backed segments (physical multicast routing handles it). (2) Use a multicast replication service appliance in the overlay (third-party solutions). (3) NSX-T 3.1+ introduced partial multicast support for specific use cases. Gateway Firewall on Edge nodes can selectively allow multicast routing between segments. Multicast support in NSX-T continues to evolve — check the specific version's release notes for current capabilities.

Q 78

What is the NSX-T Infra Hierarchy in Policy API?

Answer

The NSX-T Policy API uses a hierarchical object model under /infra (Infrastructure): Infra > Tier-0s > Tier-1s > Segments > Ports. Infra > Domains > Groups > Security Policies > Rules. Infra > Services > Service Entries. Infra > Certificates, IP Pools, Transport Zones, Edge Clusters. This hierarchy means deleting a parent object cascades to children (e.g., deleting a T1 also deletes connected segments). The infra hierarchy enables bulk configuration export/import and simplifies automation — a single API call to /infra with a nested JSON body can create a complete multi-tier network topology.

Q 79

What is NSX-T BFD configuration on Edge?

Answer

BFD (Bidirectional Forwarding Detection) on NSX Edge is configured in two places: (1) Edge Cluster Profile: controls BFD between Edge nodes for HA failover detection (BFD probe interval 300-500ms, multiplier 3-5 — failure detected in ~1-2 seconds). (2) T0 BGP Neighbor configuration: BFD enabled per BGP neighbor for fast BGP failover detection. BFD parameters: TX interval (how often to send BFD probes), RX interval (minimum expected receive interval), multiplier (missed probes before failure declared). BFD requires both ends to be configured — the physical router must also support and enable BFD for the T0 BGP session. BFD is critical for sub-second failover in active-standby deployments.

Q 80

What is NSX-T host preparation rollback?

Answer

If NSX-T host preparation fails or needs to be reversed (e.g., removing NSX-T from a cluster), the rollback process removes NSX VIBs from ESXi hosts: (1) Migrate all VMs off the host (maintenance mode). (2) In NSX Manager, remove the host from the Transport Node Profile or delete the Transport Node configuration. (3) NSX Manager sends a VIB removal command to the host. (4) Host reboots to complete VIB removal. Rollback also removes the N-VDS — all VMs connected to NSX segments must be migrated to standard port groups before rollback. In vSphere 7.0+ (VDS-based N-VDS), rollback removes the NSX configuration from VDS without full VIB removal.

Q 81

What is NSX-T and microsegmentation for containers?

Answer

NSX-T micro-segmentation for containers: NCP translates Kubernetes NetworkPolicy objects into NSX-T DFW rules at the pod level. Each pod gets a unique logical port on an NSX-T segment. DFW rules are applied to pod ports the same way as VM ports. Container-specific considerations: pods are ephemeral (short-lived), so security groups must use dynamic criteria (pod labels/namespaces) not static IPs. NCP automatically updates DFW group membership as pods start/stop. For service-to-service security: use Kubernetes NetworkPolicy for pod-level rules, and NSX-T Gateway Firewall for namespace-to-namespace or cluster-to-external security.

Q 82

What is NSX-T segment tagging?

Answer

NSX-T segments (Logical Switches) support NSX-T Tags – the same key-value metadata system used for VMs. Segment tags are used for: organizing segments in large environments (e.g., Tag: Tier=Web, Environment=Production), dynamic group membership (create groups matching segments by tag for firewall 'Applied To' scope), and API-based automation (filter segments by tag in GET requests for bulk operations). Segment tags are different from VLAN tags – they are NSX-T metadata labels, not network encapsulation identifiers. Apply tags when creating or editing a segment via NSX Manager UI or API.

Q 83

What is NSX-T trunk port vs access port for Edge uplinks?

Answer

NSX-T Edge uplinks require trunk ports on physical switches because: Edge uplinks carry tagged VLAN traffic for multiple purposes – T0 uplink VLAN (for routing to physical routers), TEP VLAN (for overlay tunnels to hosts). Physical switch ports connected to the ESXi host NICs (that host Edge VMs) must be configured as 802.1Q trunk ports allowing: the TEP VLAN, all T0/T1 uplink VLANs, and the Edge management VLAN. Access ports (single untagged VLAN) are not suitable. VLAN configuration error on the physical switch is one of the most common causes of Edge connectivity failures – verify with 'show interfaces trunk' on the physical switch and compare allowed VLANs with the Edge N-VDS VLAN configuration.

Q 84

What is NSX-T Network Topology view?

Answer

NSX Manager provides a Network Topology visualization (Networking > Network Topology) that shows a graphical map of: T0/T1 gateways, connected segments, segment interconnections, Edge clusters, and connected workloads. The topology view helps administrators: quickly understand the network design, identify misconfigured or disconnected components (shown in red/orange), trace connectivity paths visually, and onboard new team members with a visual understanding of the environment. The topology view is read-only – all configuration changes are made via the standard UI. For very large environments, the topology view can be filtered by T0 or T1 gateway to reduce visual complexity.

Q 85

What is NSX-T multi-TEP per host?

Answer

NSX-T supports multiple TEP IP addresses on a single ESXi host — one TEP per N-VDS uplink (for ECMP-enabled deployments). With ECMP enabled on the host TEP, overlay traffic from the host can be load-balanced across multiple TEPs (and thus multiple physical uplinks) simultaneously. This provides both bandwidth aggregation and redundancy for overlay traffic. Multi-TEP configuration: in the Uplink Profile, configure multiple active uplinks; each uplink gets its own TEP IP from the TEP pool. The TEP ECMP mode uses IP hash (source/destination IP of the Geneve outer header) for flow-based load balancing. Multi-TEP is especially beneficial on hosts with 2x25GbE or 4x10GbE NICs dedicated to overlay traffic.

Q 86

What is NSX-T overlay network troubleshooting with Wireshark?

Answer

For deep overlay troubleshooting using Wireshark: (1) Capture traffic on the TEP VMkernel interface of the source or destination ESXi host. (2) In Wireshark, apply filter: `udp.port == 6081` (Geneve traffic). (3) Wireshark 3.x+ natively decodes Geneve — the inner Ethernet frame is visible in the packet dissection. (4) Verify: outer IP headers (TEP source = source host TEP IP, TEP destination = destination host TEP IP), Geneve VNI (identifies the segment), inner frame (actual VM traffic). (5) Check for fragmented Geneve packets (indicates MTU issue — inner packet + Geneve overhead exceeds physical MTU). Packet captures on the TEP interface capture all overlay traffic passing through that host's TEP.

Logical Routing – Tier-0 & Tier-1 Gateways

32 Q&A;

Q 87

What is a Tier-0 Gateway in NSX-T?

Answer

A Tier-0 (T0) Gateway is the NSX-T router that connects the NSX overlay network to the physical network (North-South routing). It runs on NSX Edge nodes and peers with physical routers via BGP or OSPF to exchange routes. T0 supports active-active or active-standby HA modes for the uplink. It handles traffic entering and leaving the NSX fabric from external networks. The T0 gateway is the border between the NSX overlay world and the physical network, and it is the only NSX component that talks to physical routers.

Q 88

What is a Tier-1 Gateway in NSX-T?

Answer

A Tier-1 (T1) Gateway is an NSX-T router that provides routing for VMs and workloads within the NSX overlay. T1 gateways connect to segments (logical switches) on their downlinks and connect to a T0 gateway on their uplink. T1 handles East-West routing between segments attached to it, and North-South traffic is forwarded up to the T0. T1 is typically created per tenant, application, or VPC to provide routing isolation. T1 gateways can be centralized (running only on Edge nodes) or distributed (routing done on each ESXi host for best performance).

Q 89

What is the difference between Centralized and Distributed T1?

Answer

Centralized T1 (also called Stateful Active/Standby): All routing is performed on the active Edge node. Traffic from any host must traverse the Edge node for routing – not optimal for East-West traffic. Required when stateful services (NAT, load balancer, stateful firewall) are needed on the T1. Distributed T1: Routing between segments is performed directly on each ESXi host (in the distributed dataplane). Traffic between two VMs on the same host never leaves the host for routing – maximum performance. Services requiring statefulness (NAT) are still offloaded to Edge. Most T1 gateways should be configured as Distributed with Edge cluster for stateful services.

Q 90

What is the role of Distributed Router (DR) in NSX-T?

Answer

The Distributed Router (DR) is the data-plane component of a T1/T0 gateway that runs on every transport node (ESXi host) in the overlay. The DR handles intra-host and inter-host routing at line rate without sending traffic to a central device. When VM-A (on Segment-A, 10.0.1.x) sends traffic to VM-B (on Segment-B, 10.0.2.x), the DR on the source host routes the packet locally. The DR component is the foundation of NSX-T's high-performance distributed routing – eliminating hair-pinning through physical routers or central VMs for East-West traffic.

Q 91

What is a Service Router (SR) in NSX-T?

Answer

The Service Router (SR) is the component of a T0/T1 gateway that runs on NSX Edge nodes and handles stateful services that cannot be distributed: NAT (Network Address Translation), Edge Firewall, Load Balancer, VPN (IPsec/L2VPN), BGP/OSPF peering with physical networks, and DHCP relay/server. When T1 is configured with NAT or LB, an SR is instantiated on the Edge cluster. The DR handles fast-path routing on each host; the SR handles stateful services on Edge. Together they form the complete NSX-T gateway pipeline.

Q 92

How does BGP work with NSX-T Tier-0?

Answer

T0 gateways peer with physical routers via BGP to exchange routes. Configuration: Add BGP neighbors (physical router IPs) on the T0 uplink interface, configure AS numbers (T0 ASN and neighbor ASN), and enable route redistribution (Connected, T1 routes, Static routes) into BGP. The T0 advertises NSX overlay subnet routes to the physical network via BGP UPDATE messages. Physical routers respond with their routes (default route or specific prefixes). For redundancy, the T0 should peer with at least two physical routers (spine switches) and use ECMP (Equal-Cost Multi-Path) for active-active uplink utilization.

Q 93

What is ECMP in NSX-T Tier-0?

Answer

ECMP (Equal-Cost Multi-Path) on T0 allows traffic to be distributed across multiple active uplinks or Edge nodes using multiple equal-cost paths simultaneously. For active-active T0 HA mode, multiple Edge nodes all advertise the same routes to physical routers via BGP. Physical routers distribute traffic across all Edge nodes using ECMP. This provides both load balancing and redundancy – if one Edge fails, traffic is redistributed to remaining nodes. ECMP requires the physical routers to support ECMP and is configured by advertising the same BGP next-hop from multiple Edge nodes.

Q 94

What is Route Redistribution in NSX-T?

Answer

Route Redistribution in NSX-T controls which routes are advertised from the T0/T1 into routing protocols (BGP/OSPF) or redistributed between routing domains. On T0, you configure which route types to redistribute into BGP: Connected Routes (T0 uplink subnets), T1 Advertised Routes (subnets from T1 gateways), Static Routes, NSX-connected segment routes. On T1, you configure which routes to advertise to T0: Connected routes (attached segments), NAT IP addresses. Proper route redistribution ensures physical routers know how to reach all NSX overlay subnets.

Q 95

What is NAT in NSX-T and what types are supported?

Answer

NSX-T supports multiple NAT types on T0 and T1 gateways: SNAT (Source NAT) – translates VM private IPs to a public IP for outbound traffic; DNAT (Destination NAT) – maps a public IP to a private VM IP for inbound traffic (publishing services); Reflexive NAT (No-SNAT / No-DNAT) – explicitly bypasses NAT for specific traffic; NAT64 – translates between IPv6 and IPv4 addresses. NAT rules have priority and are evaluated top-down. NAT runs on the SR (Service Router) on Edge nodes – so NAT traffic must traverse Edge nodes. Distributed NAT was introduced in NSX-T 3.0 for better performance.

Q 96

What is a Static Route in NSX-T?

Answer

Static routes in NSX-T are manually configured routes on T0 or T1 gateways pointing traffic to specific next-hop IP addresses. Use cases: adding routes to subnets not reachable via BGP/OSPF, pointing null routes for blackholing specific prefixes, routing to VPN-connected networks, or for environments where dynamic routing is not used. T0 static routes are propagated to T1 gateways if configured in route redistribution. Static routes on T1 can be advertised to T0 if T1 route advertisement is enabled. BFD can be associated with static routes for automatic route removal on next-hop failure.

Q 97

What is the Linked Port between T0 and T1?

Answer

When a T1 gateway is connected to a T0 gateway, NSX-T automatically creates an internal transit subnet (169.254.0.0/28 by default – RFC 5735) between them. This is called the Linked Port connection. The T1 uses this transit link to send its advertised subnets as connected routes to the T0. The transit subnet is hidden from administrators and managed automatically by NSX-T. Each T1-T0 pair gets a unique transit subnet. This design allows multiple T1 gateways to connect to the same T0 without IP address conflicts on the transit link.

Q 98

What is NSX-T Forwarding Policy (Policy-Based Routing)?

Answer

NSX-T supports Gateway Policies (Policy-Based Routing / PBR) on T0 and T1 gateways that allow traffic matching specific criteria (source IP, destination IP, protocol) to be forwarded to a specific next-hop rather than using the routing table. This is used for: traffic steering to specific firewalls or service appliances, forcing specific application traffic through a VPN tunnel, or overriding the default routing decision for certain flows. Gateway Policies are evaluated before the routing table and take priority over static/dynamic routes.

Q 99

What is a Segment connected to T1 and how does routing work?

Answer

When a Segment is connected to a T1 gateway, the T1 creates a logical interface (gateway IP) on that segment – typically the default gateway IP for VMs on that segment (e.g., 10.0.1.1/24 for a 10.0.1.0/24 segment). VMs use this gateway IP as their default gateway. When a VM sends traffic to another subnet, the DR component on the ESXi host routes the packet to the T1's DR, which routes it to the correct destination. The T1 automatically advertises connected segment routes to the T0. Administrators set the gateway IP when connecting the segment to the T1 in NSX Manager.

Q 100

What is VRF (Virtual Routing and Forwarding) in NSX-T?

Answer

NSX-T supports VRF-Lite on Tier-0 gateways to provide routing isolation between multiple tenants or network domains sharing the same T0 gateway infrastructure. Each VRF has its own routing table, BGP sessions, and uplink interfaces – traffic from different VRFs is completely isolated even though they share the same T0 (and Edge nodes). VRF-T0 gateways (VRF instances created on a parent T0) share the Edge cluster with the parent T0 but maintain separate routing tables and BGP peerings. This is used in multi-tenant environments where different business units need isolated routing.

Q 101

What happens to routing when an Edge node fails in NSX-T?

Answer

In active-standby T0/T1: the standby Edge detects the active Edge failure via BFD (typically within 1 second), takes over the VIP/active role, and re-establishes BGP sessions with physical routers. Traffic is briefly interrupted during failover. In active-active T0 with ECMP: if one Edge node fails, physical routers detect the failure (via BFD or BGP hold-down timer) and remove that Edge's paths from ECMP. Traffic is immediately redistributed to remaining active Edge nodes. Active-active provides faster failover and better performance but requires ECMP support on physical routers.

Q 102

What is the Connected Segments Advertisement on T1?

Answer

T1 gateways have a setting called 'Route Advertisement' that controls which routes the T1 advertises to the connected T0 gateway. Options: All Connected Segments and Service Ports (advertises all segments attached to T1), All Static Routes, All NAT IPs, All LB VIPs (advertise load balancer virtual IPs to T0 for external access), All IPsec local endpoints, DNS forwarder IPs. By default, T1 only advertises its connected segments. For external services (NAT, LB), you must explicitly enable the corresponding advertisement options so that the T0 (and subsequently physical routers via BGP) know how to reach those IPs.

Q 103

What is OSPF support in NSX-T Tier-0?

Answer

NSX-T Tier-0 gateways support OSPF (Open Shortest Path First) as an alternative to BGP for dynamic routing with physical networks. OSPF on T0 is configured with: area ID (backbone area 0 or other areas), OSPF interface type (Point-to-Point for direct router connections, or Broadcast for multi-access networks), hello/dead intervals, and redistribution of connected routes and T1 routes into OSPF. BGP is generally preferred for data center environments (better policy control, scalability), but OSPF is useful for integration with existing campus/WAN networks that use OSPF natively.

Q 104

What is the NSX-T T0 active-standby vs active-active mode?

Answer

Active-Standby T0: one Edge node runs the active SR (handles all routing/services), the other is standby (takes over on active failure via BFD detection). Simpler configuration, supports all stateful services (NAT, LB, VPN). Failover causes brief traffic interruption during SR takeover. Active-Active T0: multiple Edge nodes all run active SRs simultaneously. Traffic is distributed across Edge nodes via ECMP on physical routers (requires BGP ECMP support on physical routers). Better throughput and redundancy – no single point of failure. Limitation: stateful services (NAT, LB) cannot be fully active-active (sessions are tied to one SR). Best practice: active-active for routing throughput, active-standby where stateful services are needed.

Q 105

What is the T0/T1 route advertisement?

Answer

Route advertisement controls which routes a T1 gateway sends up to its connected T0 gateway. Without advertisement, the T0 does not know about subnets behind the T1 and cannot route traffic to them. T1 can advertise: All Connected Segments and Service Ports (all subnets directly attached to T1), All Static Routes, All NAT IPs (for external access to NATted services), All LB VIPs (load balancer virtual IPs for external access), All IPSec Local Endpoints, All DNS Forwarder IPs. Enable the appropriate advertisement options based on what services are configured on the T1. Missing route advertisement is a common cause of connectivity failures.

Q 106

What is the NSX-T Prefix Lists and Route Maps?

Answer

Prefix Lists define sets of IP prefixes (subnets) used for route filtering — specify which routes to accept or deny in BGP. Route Maps extend prefix filtering with policy actions: set BGP attributes (local preference, MED, AS path prepend, community) on matched routes. Example use case: on T0 BGP configuration, use a Route Map to set higher local preference on routes learned from the primary physical router (to prefer it over secondary), or to add an AS path prepend to routes advertised to a secondary ISP (to influence inbound traffic). Prefix Lists and Route Maps are configured in NSX-T under T0 Gateway > Routing > BGP Neighbors.

Q 107

What is the NSX-T DNS Forwarder?

Answer

NSX-T includes a DNS Forwarder service that can run on T0 or T1 gateways (on Edge nodes). The DNS Forwarder receives DNS queries from VMs in the overlay and forwards them to upstream DNS servers. This avoids requiring VMs to have direct network access to corporate DNS servers — they send queries to the T1's DNS forwarder IP instead. Zone-based forwarding allows different DNS zones to be forwarded to different DNS servers (e.g., internal .company.com queries to corporate DNS, all others to 8.8.8.8). The DNS Forwarder IP must be advertised from T1 to T0 via route advertisement.

Q 108

What is hairpinning in NSX-T and how to avoid it?

Answer

Hairpinning occurs when VM-to-VM traffic within the overlay must traverse a central device (Edge node or physical router) for routing, instead of being handled locally on the host. In NSX-T, hairpinning is avoided by: using Distributed Routing (T1 distributed mode) – inter-segment routing happens on the source ESXi host's DR, never leaving the host for intra-cluster traffic. Hairpinning still occurs for: North-South traffic (must traverse Edge T0/T1 SR), NAT/LB traffic (handled on Edge SR), and traffic between T1s connected to the same T0 (must go through T0 SR on Edge if T1s are on different Edge clusters). Design topology to minimize cross-Edge routing for East-West traffic.

Q 109

What is NSX-T Routing Policy for Tier-1?

Answer

T1 Routing Policy (Gateway Policy on T1) allows policy-based routing on the T1 gateway – forward specific traffic to a designated next-hop instead of the normal routing table. Example: all HTTP/HTTPS traffic from App-Segment should go through a security inspection service (not directly to T0). Configure a Gateway Policy on T1 with a match rule (source: App-Segment, destination: any, service: HTTP/HTTPS) and an action (redirect to Service VM IP). T1 Routing Policies are evaluated before the routing table lookup, allowing traffic steering for security inspection or WAN optimization services inline.

Q 110

What is Inter-VRF routing in NSX-T?

Answer

Inter-VRF routing in NSX-T allows selective route leaking between VRF instances on a T0 gateway. By default, each VRF has a completely isolated routing table – VMs in VRF-A cannot communicate with VMs in VRF-B. For shared services (DNS, AD, backup servers) that need to be accessible from all VRFs, configure route leaking: export specific routes from a shared-services VRF into other VRFs. Route leaking is configured via Route Maps and Prefix Lists on the VRF BGP configuration. This maintains routing isolation between tenants while allowing controlled access to shared infrastructure services.

Q 111

What is NSX-T Service Account vs User Account?

Answer

User Accounts in NSX-T are for human administrators — created locally in NSX Manager (System > User Management > Users) or mapped from Active Directory via vIDM. They use username/password authentication and support MFA via vIDM. Service Accounts (Principal Identities) are for machine-to-machine API access — they use X.509 certificate authentication (no password), are assigned specific RBAC roles, and are used by automation tools (Terraform, vRA, Kubernetes NCP). Service accounts should be created per automation tool, following least-privilege, and certificates should be rotated periodically. Audit logs distinguish between human and service account actions.

Q 112

What is NSX-T DFW and vSphere Tags?

Answer

NSX-T DFW groups can be defined using vSphere Tags applied to VMs in vCenter. When a vCenter Tag is applied to a VM, NSX-T reads the tag via the Compute Manager integration and automatically includes the VM in matching NSX-T groups. This enables a seamless workflow: vSphere/vRA/Terraform applies vCenter tags during VM provisioning □ NSX-T DFW automatically applies security policies to tagged VMs. vSphere Tags are particularly useful for organizations that already use vCenter tagging for inventory management — they can reuse the same tags for both vSphere operations and NSX-T security, reducing administrative overhead.

Q 113

What is NSX-T REST API rate limiting?

Answer

NSX-T Manager API has rate limiting to protect the management plane from overload: default limit is ~100 API requests per second per user. If rate limits are exceeded, the API returns HTTP 429 (Too Many Requests). For automation scripts making many rapid API calls: implement exponential backoff retry logic (wait, then retry on 429), use batch API calls where available (create multiple objects in one API call), and spread API calls over time. For large environments with multiple automation tools, ensure each tool uses a dedicated service account so rate limits are not shared. API rate limits are configurable by VMware support for specific use cases.

Q 114

What is DHCP on NSX-T T1 gateways?

Answer

NSX-T T1 gateways can run a DHCP Server to provide IP address management for VMs on attached segments. Configure DHCP server: create a DHCP Server Profile in NSX Manager (IP pool, lease time, DNS servers, default gateway, NTP, WINS options). Attach the DHCP Profile to the T1 gateway. When a VM on an attached segment sends a DHCP DISCOVER, the T1's DR (on the same host) handles the DHCP request locally without sending it to an Edge node (distributed DHCP). For environments with existing DHCP infrastructure, configure DHCP Relay on the T1 segment – DHCP requests are forwarded to an external DHCP server. NSX-T DHCP removes the need for dedicated DHCP appliances.

Q 115

What is NSX-T T0 uplink connectivity verification?

Answer

T0 uplink connectivity verification steps: (1) SSH to Edge node. (2) Enter T0 SR VRF context: vrf (get VRF ID from: get logical-routers). (3) Check interface IPs: get interface all. (4) Verify ARP for physical router: get arp. (5) Ping physical router: ping source . (6) Check BGP: get bgp summary (should show Established state with physical router). (7) Verify routes received: get route bgp (should show routes from physical router). (8) Check route redistribution: get route (should show T1 connected routes redistributed into T0). Common issue: VLAN misconfiguration on Edge uplink segment – verify ESXi host trunk port allows the uplink VLAN.

Q 116

What is NSX-T Gateway Firewall vs DFW for North-South?

Answer

For North-South traffic security (external to overlay), NSX-T provides two options: (1) DFW Applied to segment border: DFW rules with Applied-To = specific segment can control what traffic enters/exits a segment, but DFW runs after the packet is already inside the overlay – it cannot inspect packets before they are routed by T0. (2) Gateway Firewall on T0/T1: explicitly placed at the gateway interface, inspects all inbound/outbound traffic at the routing boundary. Better positioned for perimeter security. Best practice: use Gateway Firewall on T0 uplink for Internet/external traffic inspection (perimeter), use DFW for East-West micro-segmentation (internal), and use Gateway Firewall on T1 downlinks for application-tier perimeter control.

Q 117

What is NSX-T T1 centralized services port?

Answer

When a T1 gateway provides stateful services (NAT, LB, DHCP server), it needs a Service Port — a centralized logical interface on the Edge SR used for processing stateful flows. The Service Port acts as the entry/exit point for stateful service traffic on the T1 SR. All traffic that requires stateful processing (NAT translation, LB forwarding, VPN termination) flows through the Service Port on the Edge node. The Service Port IP is allocated from the T0-T1 transit subnet. Administrators do not directly configure Service Ports — NSX-T creates them automatically when stateful services are enabled on a T1 and the T1 is connected to an Edge cluster.

Q 118

What is NSX-T T0 peering with Spine-Leaf architecture?

Answer

In a modern Spine-Leaf data center fabric, NSX-T T0 gateways peer with Leaf switches (Top-of-Rack) via BGP. Each NSX Edge node connects its uplinks to two different Leaf switches (one per uplink) for redundancy. T0 BGP is configured with two BGP neighbors: Leaf-1 and Leaf-2. With ECMP, both BGP paths are active simultaneously — traffic is distributed across both Leaf switches. Leaf switches propagate NSX-T overlay routes to the Spine switches via the existing BGP fabric. This design provides: N-path redundancy, no single point of failure, and maximum East-West throughput for North-South ingress/egress traffic. BGP route summarization at the T0 reduces the number of prefixes advertised to the fabric.

NSX Edge Nodes & Edge Clusters

28 Q&A;

Q 119

What is an NSX Edge Node?

Answer

An NSX Edge Node is a dedicated VM or bare-metal appliance that runs the NSX Service Router (SR) component — providing stateful services that cannot be distributed across hypervisor hosts: North-South routing (connecting overlay to physical network), BGP/OSPF dynamic routing, NAT, Load Balancer, IPsec VPN, L2 VPN, Edge Firewall, and DHCP. Edge nodes have dedicated uplink interfaces connected to physical network segments (VLAN-backed) and overlay TEP interfaces for communication with host transport nodes.

Q 120

What are the Edge Node deployment sizes?

Answer

NSX Edge nodes are available in four sizes: Small (2 vCPU, 4 GB RAM — lab/testing only, no LB), Medium (4 vCPU, 8 GB RAM — production with small-scale services), Large (8 vCPU, 32 GB RAM — production with LB and higher throughput), Extra-Large (16 vCPU, 64 GB RAM — high-performance LB, large-scale deployments). Bare-metal Edge nodes provide the highest performance (hardware-accelerated forwarding). For production NSX-T deployments with load balancing, Large or Extra-Large is required. The size determines the maximum throughput and number of concurrent connections for stateful services.

Q 121

What is an Edge Cluster?

Answer

An Edge Cluster is a logical grouping of one or more NSX Edge nodes. T0 and T1 gateways are attached to an Edge cluster (not individual Edge nodes). NSX-T distributes SRs across the Edge nodes in the cluster for HA and load distribution. For active-standby T0/T1: one Edge is active (running the SR) and one is standby. For active-active T0: multiple Edges in the cluster all run SRs simultaneously with ECMP. An Edge cluster should have at least 2 Edge nodes for production (N+1 redundancy). A single cluster can have up to 10 Edge nodes.

Q 122

What interfaces does an NSX Edge Node have?

Answer

An NSX Edge VM has four network interfaces: Management Interface (eth0 / vNIC 0) – connected to management network for NSX Manager communication, SSH access, and syslog. Overlay (TEP) Interface (fp-eth0 / vNIC 1) – connected to TEP VLAN for Geneve overlay tunnels to host transport nodes. Uplink Interface 1 (fp-eth1 / vNIC 2) – connected to the physical network (VLAN segment) for North-South routing (T0 uplink). Uplink Interface 2 (fp-eth2 / vNIC 3) – second uplink for redundancy or additional routing paths. The management interface uses the regular Linux networking stack; the data interfaces (fp-eth*) use DPDK for high-performance packet processing.

Q 123

What is DPDK in NSX Edge?

Answer

DPDK (Data Plane Development Kit) is a high-performance packet processing framework used by NSX Edge nodes. The Edge data interfaces (fp-eth0, fp-eth1, fp-eth2) bypass the Linux kernel networking stack and use DPDK for line-rate packet processing. DPDK uses polling-mode drivers (PMD) – the CPU cores assigned to DPDK constantly poll for incoming packets rather than using interrupts, reducing latency and increasing throughput significantly. This is why Edge nodes require dedicated vCPU cores (NUMA-aware placement is recommended) and why the Edge VM must not be migrated to hosts with different NUMA topology.

Q 124

What is the difference between VM Edge and Bare-Metal Edge?

Answer

VM Edge (Virtual Edge): deployed as a virtual machine on ESXi hosts. Faster to deploy, flexible placement, supports vMotion for maintenance. Limited by hypervisor overhead – practical throughput is 40-100 Gbps depending on hardware. Suitable for most enterprise deployments. Bare-Metal Edge: deployed as NSX software directly on physical servers (without a hypervisor). Full access to physical NICs (SR-IOV, DPDK with NIC offload). Supports 100+ Gbps throughput for high-traffic environments (service providers, large financial institutions). Required for very high performance VPN or LB workloads. Cannot vMotion.

Q 125

How do you prepare an Edge node as a transport node?

Answer

Steps: (1) Deploy Edge VM from NSX Manager (System > Fabric > Nodes > Edge Transport Nodes > Add). (2) Configure management IP, compute manager (vCenter), ESXi host placement, and datastore. (3) Set CLI admin/root passwords. (4) After deployment, NSX Manager automatically prepares the Edge as a transport node — installing the NSX software components. (5) Configure the N-VDS on the Edge: assign uplink profile, TEP IP pool, and map physical interfaces (fp-eth0 to TEP, fp-eth1/fp-eth2 to uplinks). (6) Add the Edge to an Edge cluster. (7) Connect T0 gateway uplinks to VLAN segments on the Edge.

Q 126

What is Edge Node NUMA-aware placement?

Answer

NSX Edge nodes use DPDK for high-performance packet forwarding, which benefits from CPU and memory being on the same NUMA node. Edge nodes should be placed on ESXi hosts where: all vCPUs of the Edge VM are on the same NUMA node, the physical NICs used by the Edge are connected to the same NUMA node's PCIe bus, and sufficient memory is available on that NUMA node. Improper NUMA placement causes DPDK to access remote memory (cross-NUMA), increasing latency and reducing throughput. Configure DRS anti-affinity rules to pin Edge VMs to specific hosts for consistent performance.

Q 127

What is the NSX Edge Node fast path (fp-eth)?

Answer

The fp-eth interfaces (fp-eth0, fp-eth1, fp-eth2) are the NSX Edge data plane interfaces that use DPDK for high-performance forwarding. They bypass the Linux kernel network stack entirely. fp-eth0 is used for TEP (overlay) traffic, fp-eth1 and fp-eth2 for uplink traffic. These interfaces are visible in the Edge CLI but are managed by the NSX data plane process, not the Linux OS. You cannot configure them via standard Linux networking commands (ifconfig, ip). Configuration is done via NSX Manager (N-VDS configuration on the Edge transport node).

Q 128

What is the recommended redundancy configuration for Edge Clusters?

Answer

For production Edge cluster redundancy: minimum 2 Edge nodes per cluster (preferably 4 for active-active T0 with ECMP). Place Edge VMs on different ESXi hosts using DRS anti-affinity rules (so host failure does not take both Edges down). Place on different racks/power domains if possible. For T0 active-standby: 2 Edges (one active SR, one standby SR). For T0 active-active with ECMP: 2-4 Edges (all running active SRs). For large-scale LB deployments: 4+ Large or XL Edge nodes in the cluster to distribute load balancer instances.

Q 129

What is NSX Edge heartbeat and failure detection?

Answer

NSX Edge nodes send periodic heartbeats to NSX Manager to confirm they are alive and operational. Additionally, BFD (Bidirectional Forwarding Detection) is used between active and standby SR instances to detect data-plane failures quickly (sub-second). For T0 BGP uplinks, BFD is configured between the T0 SR and physical routers to detect link failures faster than BGP hold-down timers. Control-plane communication between Edge nodes and the CCP (Central Control Plane) uses a secure channel that also serves as a liveness indicator.

Q 130

What is the Edge TEP vs host TEP?

Answer

Both Edge nodes and ESXi hosts have TEP (Tunnel Endpoint) interfaces for Geneve overlay tunnels, but they serve different purposes: Host TEP — used for VM-to-VM overlay traffic between hosts (horizontal/East-West traffic between VMs on different hosts). Edge TEP — used for traffic between the Edge SR (handling North-South or stateful services) and the host DRs. When a VM's traffic needs to go through an Edge node (e.g., for NAT or going to the physical network), it travels via the Geneve overlay from the host TEP to the Edge TEP. Edge TEPs should be on a separate VLAN from host TEPs in some designs for traffic separation.

Q 131

What is NSX Edge Node CLI and common troubleshooting commands?

Answer

NSX Edge CLI (accessed via SSH to the Edge node) key commands: get interfaces (show all interfaces including fp-eth), get logical-routers (list SR and DR instances), vrf (enter a logical router context), get route (show routing table), get bgp neighbor summary (BGP peer status), get bfd-sessions (BFD session status), get forwarding (show forwarding entries), get arp (ARP table), ping source (connectivity test), get services (running services status). Use debug commands with caution in production — they can impact forwarding performance.

Q 132

What is the max number of Edge nodes per cluster?

Answer

NSX-T supports a maximum of 10 Edge nodes per Edge cluster. For most enterprise deployments, 2-4 Edge nodes per cluster is sufficient. If more capacity is needed (e.g., very high LB throughput, hundreds of VPN tunnels), multiple Edge clusters can be created — each T0/T1 gateway is attached to one cluster. Different application tiers or tenants can have their own dedicated Edge clusters for performance isolation and security boundaries. The NSX Configuration Maximums guide specifies current limits for each NSX-T/NSX version.

Q 133

How does the NSX Edge uplink connect to the physical network?

Answer

NSX Edge uplinks connect to the physical network via VLAN-backed segments (created in a VLAN transport zone). The Edge uplink interface (fp-eth1/fp-eth2) is mapped to these VLAN segments in the N-VDS configuration. The physical ESXi host's NICs (connected to the Edge VM) must be on trunk ports that allow the uplink VLANs. The T0 gateway's uplink interfaces are configured with IP addresses on the uplink VLAN segments, and BGP neighbors are configured pointing to the physical router's IP on the same uplink VLAN. This creates a routed connection from NSX-T to the physical network.

Q 134

What is the NSX Edge configuration validation?

Answer

After deploying an NSX Edge node and before adding it to an Edge cluster, validate: (1) Edge management IP is reachable from NSX Manager (ping test). (2) Edge appears in NSX Manager > System > Fabric > Nodes > Edge Transport Nodes with status 'Up'. (3) TEP configuration — Edge has TEP IP assigned from the TEP pool. (4) Overlay connectivity — Edge TEP can reach host TEPs (test with Traceflow or ping from Edge CLI). (5) Uplink connectivity — Edge uplink interfaces can reach physical router IPs (ping from Edge CLI: vrf ; ping source). (6) BGP sessions establish after T0 is configured and uplinks are connected.

Q 135

What is the NSX Edge Node maintenance mode?

Answer

When an Edge node needs maintenance (upgrade, hardware replacement), put it in maintenance mode via NSX Manager: System > Fabric > Nodes > Edge Transport Nodes > [Edge] > Actions > Enter Maintenance Mode. In active-standby T0/T1: the standby Edge takes over (brief traffic interruption during failover). In active-active T0 with ECMP: the failing Edge is removed from ECMP rotation; physical routers redirect traffic to remaining active Edges. After maintenance, exit maintenance mode: the Edge rejoins the cluster, BGP sessions re-establish, and DRS may move it back to active routing role. Always verify cluster health after exiting maintenance mode.

Q 136

How does Edge node CPU pinning work?

Answer

NSX Edge VM performance benefits from dedicated CPU resources — the DPDK data plane uses polling-mode drivers that require dedicated vCPU cores for low-latency forwarding. CPU pinning configuration: in vCenter, configure VM CPU affinity for the Edge VM to pin its vCPUs to specific physical cores. Ensure the pinned cores are on the same NUMA node as the Edge's physical NICs. Disable CPU hot-add on Edge VMs (incompatible with DPDK). Enable hardware CPU reservation (reserve 100% of Edge vCPUs) so Edge vCPUs are never over-committed or stolen by the ESXi scheduler. NSX documentation provides detailed NUMA-aware CPU/NIC topology guidelines for each Edge VM size.

Q 137

What is an NSX Edge Cluster profile?

Answer

An Edge Cluster Profile defines HA settings for the Edge cluster: BFD probe interval (how often BFD probes are sent between cluster members — lower = faster failover but more overhead), BFD timeout multiplier (how many missed probes before declaring a node failed), and standby relocation timeout (how long to wait before relocating the standby SR to another Edge node after failover). Default BFD interval is 500ms with multiplier 3 (failure detected in ~1.5 seconds). For faster failover (e.g., trading or financial applications), reduce BFD interval to 300ms. Modify Edge Cluster Profile in NSX Manager: System > Fabric > Nodes > Edge Clusters > [cluster] > Edit Profile.

Q 138

What are NSX Edge uplink types?

Answer

NSX Edge uplinks connect to the physical network via VLAN-backed segments. Uplink types: External Uplinks (fp-eth1, fp-eth2) – connected to T0 gateway uplink VLAN segments for routing/BGP to physical routers. Internal Uplinks (typically fp-eth0) – connected to TEP VLAN segment for overlay tunnel traffic (Geneve encapsulation to/from host transport nodes). Some designs use a dedicated management segment for Edge management (eth0). For redundancy, use two external uplinks (fp-eth1 to spine switch 1, fp-eth2 to spine switch 2) and configure T0 with BGP peers on both, using ECMP or active-standby between them.

Q 139

What is NSX-T RBAC integration with Active Directory?

Answer

NSX-T RBAC integrates with Active Directory via vIDM (VMware Identity Manager, now Workspace ONE Access). Configuration: register vIDM with NSX Manager (System > User Management > Authentication Providers). Import AD groups/users from vIDM and assign NSX-T roles (Enterprise Admin, Network Admin, Security Admin, Auditor). Users authenticate with AD credentials via SSO – no separate NSX-T local accounts needed. Benefits: centralized access management, AD password policies apply, AD group membership changes automatically update NSX-T permissions. When an employee leaves, disabling their AD account immediately revokes NSX-T access.

Q 140

What is NSX-T DFW draft mode?

Answer

NSX-T DFW Draft Mode allows administrators to create and modify firewall policies in a draft (staging) area without immediately applying them to workloads. This enables: change review workflow – senior security admin can review proposed rules before publishing, testing – validate that new rules have correct logic without affecting production, rollback – discard draft changes without any production impact, and compliance – get change approval before implementation. Draft Mode is configured per DFW policy section. When ready, publish the draft – rules are pushed to all transport nodes simultaneously. Draft Mode is especially valuable in change-controlled environments requiring 4-eyes approval for firewall changes.

Q 141

What is NSX-T Manager disk space management?

Answer

NSX Manager disk space issues are a common operational concern: NSX Manager stores audit logs, flow logs, and configuration history which can fill disk partitions. Monitor: /var/log partition (syslog, audit logs), /common (NSX Manager database), /tmp. Alerts: NSX Manager generates Critical alarm when disk usage exceeds 80%. Management: configure log rotation (automatic in NSX Manager), configure remote syslog to offload logs to external server, use the 'clear' commands via NSX Manager CLI (SSH to manager: clear log) for immediate relief. For persistent disk space issues, increase the disk size of the NSX Manager VM (requires appliance resize procedure documented by VMware). Always monitor disk trends via Aria Operations.

Q 142

What is NSX-T Edge VM anti-affinity?

Answer

NSX Edge VM anti-affinity rules in vSphere prevent multiple Edge nodes from the same Edge cluster from running on the same ESXi host. If two Edge nodes are on the same host and that host fails, both Edge nodes fail simultaneously – eliminating redundancy. Configuration: create a VM-VM Anti-Affinity DRS rule in vCenter for each Edge cluster (keep Edge-1 and Edge-2 apart). NSX Manager does not create this rule automatically – it must be manually configured by the vSphere administrator. For 4-node Edge clusters: create anti-affinity rules keeping all 4 Edge VMs on separate hosts. This is a critical production requirement and is frequently missed during initial deployments.

Q 143

What is NSX Edge Cluster high availability modes?

Answer

NSX Edge Cluster HA modes affect how T0/T1 gateways distribute SR instances: (1) Active-Standby per T0: one Edge in cluster is active (holds all SR instances for that T0), other is standby. Failover on active failure. Simple but no load distribution. (2) Active-Active per T0 (ECMP): all Edges in cluster are active for a T0; each hosts an SR instance; physical routers distribute traffic via ECMP. Maximum throughput. (3) Per-T1 distribution: different T1 SRs are placed on different Edge nodes in the cluster (NSX auto-distributes or admin can pin T1 SRs). Provides load balancing across Edge nodes for multiple T1 gateways sharing the same Edge cluster.

Q 144

What is NSX-T Edge node management interface security?

Answer

NSX Edge node management interface (eth0) security hardening: restrict SSH access to management network IPs only (configure firewall rules or ACLs), disable password-based SSH login and use SSH key-based authentication, change the default SSH admin credentials immediately after deployment, configure NTP for accurate log timestamps, forward Edge syslog to SIEM for audit trail, apply NSX-T Edge patches promptly via the NSX Upgrade Coordinator, and monitor Edge management interface for unusual connections (brute-force SSH attempts). The management interface should be on an isolated management VLAN not accessible from VM workloads or the internet – only from the management jump host/bastion server.

Q 145

What is NSX-T Edge node OVF deployment options?

Answer

NSX Edge OVF deployment options via NSX Manager (System > Fabric > Nodes > Edge Transport Nodes > Add): Form Factor (Small/Medium/Large/XL), Compute Manager and Cluster (which ESXi cluster to deploy on), Datastore (storage for Edge VM disk), IP Assignment (static or DHCP for management), Management Network (port group for management interface), Node Settings (admin/root passwords, SSH key), and NSX Configuration (uplink profile, TEP IP pool, transport zone membership). Alternatively, deploy Edge OVA manually via vCenter and then register it with NSX Manager as an Edge transport node. NSX Manager-based deployment is preferred – it automates both VM deployment and NSX transport node configuration in one step.

Q 146

What is NSX-T Edge SR migration?

Answer

When an active Edge SR fails (Edge node crash or maintenance), NSX-T can migrate the SR to another Edge node in the cluster – this is called SR migration. For active-standby T0/T1: the standby Edge already has a pre-configured standby SR that automatically takes over (no migration needed – it is instantaneous promotion). For T1 SRs: if the active Edge node is removed from the cluster or fails permanently, NSX Manager migrates the T1 SR to another available Edge node in the cluster. SR migration may cause brief traffic interruption (connections are re-established). After SR migration, verify BGP sessions and VPN tunnels are re-established on the new Edge node.

NSX-T Distributed Firewall (DFW)

29 Q&A;

Q 147

What is NSX-T Distributed Firewall (DFW)?

Answer

NSX-T Distributed Firewall (DFW) is a stateful firewall enforced at the vNIC (virtual NIC) level of every VM in the NSX-T fabric. Unlike traditional firewalls that are deployed as dedicated appliances in the traffic path, DFW runs as a kernel module on every ESXi host. Every packet entering or leaving a VM's vNIC is inspected against DFW rules. This provides micro-segmentation — fine-grained security policies per VM, per workload, or per application tier — without hairpinning traffic through a central appliance.

Q 148

How does DFW micro-segmentation work?

Answer

Micro-segmentation with DFW works by defining security policies based on workload identity (groups/tags) rather than IP addresses or VLANs. Example: Instead of allowing all traffic on VLAN 100 (web tier), you create a group 'Web-Servers' (all VMs tagged as web) and a group 'App-Servers' (all VMs tagged as app), then create a DFW rule: Allow Web-Servers to App-Servers on port 8080. VMs tagged appropriately are automatically included in groups. DFW rules follow VMs across vMotion and are automatically updated when VMs are added/removed from groups.

Q 149

What is the DFW rule structure in NSX-T?

Answer

A DFW rule has: Name, Source (Groups, Any, or specific IPs), Destination (Groups, Any, or specific IPs), Services (TCP/UDP ports, named services), Profiles (L7 App ID, User Identity, URL categories), Action (Allow, Drop, Reject, Redirect to service), Applied-To (which workloads the rule applies to — default is all DFW-protected workloads), Direction (In/Out/In-Out, from the workload's perspective), and IP Protocol (IPv4, IPv6, IPv4/IPv6). Rules are organized in Policies, which are evaluated top-down by sequence number.

Q 150

What is the difference between Allow, Drop, and Reject in DFW?

Answer

Allow — the packet is permitted to pass through. Drop — the packet is silently discarded; the sender receives no response (connection times out). Reject — the packet is discarded AND the sender receives a TCP RST (for TCP) or ICMP Port Unreachable (for UDP) response, making the connection fail immediately rather than timing out. Drop is preferred for security (no information leakage) but causes slow application timeouts. Reject is better for user experience (immediate error) but reveals that a firewall is present. Default rule (last rule) should always be Drop to block anything not explicitly allowed.

Q 151

What is the Default DFW Policy in NSX-T?

Answer

NSX-T DFW has a Default Policy (also called Default Layer 3 Policy) that contains the default rules evaluated last after all other policies. By default, the Default Policy has one rule: 'Default Allow' – which permits all traffic. In a Zero-Trust security model, the default rule should be changed to 'Default Drop' to block all traffic not explicitly permitted. However, before changing the default to Drop, ensure all required communication flows have been explicitly allowed, otherwise you may disrupt all VM communication. NSX Intelligence can help discover required flows before changing the default.

Q 152

What are Security Groups in NSX-T?

Answer

Security Groups (called Groups in NSX-T Policy API) are dynamic collections of workloads used as source/destination in DFW rules. Groups can include: VM Name patterns (regex matching), Tags (key-value pairs applied to VMs), IP addresses/CIDR ranges, Segment/Port, Active Directory User/Group (with Identity Firewall), Segment MAC addresses, and nested groups (groups of groups). The power of groups is their dynamic nature – when a VM is tagged or named to match a group's criteria, it is automatically included in the group and subject to all firewall rules referencing that group.

Q 153

What is NSX-T Identity Firewall (IDFW)?

Answer

Identity Firewall (IDFW) extends NSX-T DFW to enforce firewall rules based on Active Directory (AD) user identity rather than just IP addresses. When a user logs into a VM (RDP/VDI/physical machine), the NSX-T Guest Introspection framework (or VMware Tools) detects the user login and maps the user's AD identity to the VM's IP. DFW rules can then specify AD users/groups (e.g., 'Finance-Group □ DB-Servers: Allow SQL') – ensuring only authenticated finance users can reach database VMs. IDFW is critical for VDI environments where multiple users may share the same floating IPs.

Q 154

What is the DFW policy hierarchy in NSX-T?

Answer

NSX-T DFW organizes rules in a hierarchical policy structure: (1) Ethernet Policies (Layer 2 – MAC address based rules), evaluated first. (2) Emergency Policies (highest priority L3 rules – for immediate response to security incidents). (3) Infrastructure Policies (for management traffic – NSX Manager, vCenter communication). (4) Environment Policies (between major environment tiers – Production, Dev, Test isolation). (5) Application Policies (micro-segmentation between application tiers). (6) Default Policy (catch-all). Higher categories take precedence over lower ones. This hierarchy allows security teams to organize rules by purpose and responsibility.

Q 155

How does DFW handle East-West traffic?

Answer

DFW inspects all East-West (VM-to-VM) traffic directly on the ESXi host where the VMs run. When VM-A sends a packet to VM-B on the same host: the packet exits VM-A's vNIC □ DFW inspects the packet against outbound rules for VM-A □ the packet is internally switched to VM-B's vNIC □ DFW inspects it against inbound rules for VM-B. All of this happens in the ESXi kernel without the packet leaving the physical host. For cross-host traffic, DFW on the source host checks outbound rules, and DFW on the destination host checks inbound rules. Zero traffic hairpins through a central firewall appliance.

Q 156

What is NSX-T L7 Application Identification in DFW?

Answer

NSX-T DFW supports Layer 7 application identification – it can identify the application protocol of a connection regardless of port number. For example, detecting HTTP traffic on a non-standard port, identifying specific cloud applications (Office 365, Salesforce, AWS), or detecting tunneling protocols (SSH over port 443). L7 App ID is configured via Context Profiles in DFW rules. NSX-T uses a combination of deep packet inspection (DPI) and machine-learning signatures to classify flows. L7 rules require the Distributed Firewall Advanced edition and increase CPU overhead on ESXi hosts.

Q 157

What is Service Insertion in NSX-T DFW?

Answer

Service Insertion (also called Partner Service Insertion or NSX-T Service Insertion) allows traffic matched by a DFW rule to be redirected to a third-party security appliance (IDS/IPS, next-gen firewall, malware sandbox) for deep inspection, instead of being handled by the NSX DFW alone. Examples: redirect all VM traffic to Trend Micro Deep Security for malware scanning, or redirect specific traffic to Palo Alto VM-Series firewall for NGFW inspection. Service Insertion uses the NSX-T Service Insertion API and integrates partner solutions via the VMware Partner Security Framework.

Q 158

What is NSX-T Endpoint Protection?

Answer

NSX-T Endpoint Protection (formerly Guest Introspection) allows third-party security solutions to inspect VM activity at the hypervisor level without installing an agent in every VM. A partner security VM (SVM – Service Virtual Machine) runs on each ESXi host. The NSX-T Guest Introspection framework hooks into VM I/O (file system events, network events) and forwards them to the SVM for analysis. Supported partners: Trend Micro, Symantec, McAfee. This agentless security model is particularly valuable for VDI environments where installing agents in hundreds of desktop VMs is impractical.

Q 159

How do you troubleshoot DFW rule issues in NSX-T?

Answer

Troubleshooting DFW issues: (1) NSX Manager > Security > Distributed Firewall > Filter/trace tool – test which rule would apply to specific source/destination/port. (2) Traceflow tool (NSX Manager > Tools > Traceflow) – inject a test packet and trace its path through the network/firewall. (3) VM-level rule verification – on ESXi host: summarize-dvfilter (shows DFW filter instances per VM). (4) View DFW rules on a host: vsipioctl -i getrules. (5) Check DFW statistics: vsipioctl stats -d . (6) Review NSX Manager audit logs for recent rule changes. (7) Use NSX Intelligence to visualize actual traffic flows and identify blocked flows.

Q 160

What is Distributed Firewall vs Edge Firewall in NSX-T?

Answer

Distributed Firewall (DFW): stateful firewall running as a kernel module on every ESXi host, enforced at each VM's vNIC. Protects East-West traffic between VMs within the NSX fabric. Zero performance impact from hair-pinning. Cannot inspect North-South traffic from external sources before it enters the overlay. Edge Firewall (Gateway Firewall): stateful firewall running on NSX Edge nodes on T0/T1 gateways. Inspects North-South traffic entering or leaving the overlay fabric. Can enforce perimeter security policies. Centralized – traffic must pass through Edge nodes. Best practice: use both – DFW for East-West micro-segmentation, Edge Firewall for North-South perimeter security.

Q 161

What is NSX-T Firewall rule compliance and auditing?

Answer

NSX-T provides policy auditing capabilities: All configuration changes are logged in the NSX Manager audit log (System > Logs > Audit Logs) – recording who made what change, when, and from which IP. DFW rules can be tagged with metadata (description, owner, ticket number). NSX Manager integrates with vRealize Log Insight (Aria Log Insight) for centralized log management and security event correlation. For compliance (PCI-DSS, HIPAA, SOC2), NSX-T Distributed Firewall rules can be exported and reviewed. NSX Intelligence provides a visual map of allowed/blocked flows for compliance evidence.

Q 162

What is the DFW capacity limit in NSX-T?

Answer

NSX-T DFW capacity limits (per NSX Manager cluster, approximate for NSX-T 3.x/4.x): Maximum DFW rules: 100,000 (across all sections/policies). Maximum Groups: 10,000 static groups, 10,000 dynamic groups. Maximum rule sections: 10,000. Maximum logical ports protected by DFW: 100,000. Maximum members per group: 500 static IP/MAC, unlimited dynamic (tag-based). Per-host limits: the ESXi host DFW filter can handle up to 10,000 rules per host. For large environments approaching these limits, use hierarchical policies, consolidate rules using groups, and consider NSX-T Intelligence for rule optimization recommendations. Check VMware Configuration Maximums for version-specific limits.

Q 163

What is DFW Stateful Firewall behavior?

Answer

NSX-T DFW is a stateful firewall — it tracks the state of network connections (TCP three-way handshake, UDP flows, ICMP echo/reply pairs). For an established TCP connection, DFW allows return traffic automatically without requiring a separate rule for the return direction. DFW maintains a connection state table per VM vNIC (not per host). This means: a rule 'Allow Web-VMs to DB-VMs on port 3306' also implicitly allows the return traffic (DB response to Web) without needing a separate rule. UDP and ICMP flows are tracked with configurable idle timeouts. Stateful behavior is critical for ensuring correct application communication while maintaining security.

Q 164

What is a DFW Exclusion List?

Answer

The DFW Exclusion List is a configurable list of VMs that are exempted from all DFW enforcement. VMs on the exclusion list send and receive traffic without any DFW inspection. Use cases: NSX Manager VMs themselves (to prevent accidental lockout), vCenter Server, other management infrastructure VMs (backup proxies, monitoring agents), and VMs with complex networking that DFW cannot handle correctly (e.g., some NIC bonding configurations). Add VMs to exclusion list: NSX Manager > Security > Distributed Firewall > Exclusion List. Exclude only VMs that absolutely need it — every excluded VM is a potential security gap. Document all exclusions with justification.

Q 165

What is NSX-T Firewall Rule scope (Applied To)?

Answer

The 'Applied To' field in a DFW Policy determines which workloads the policy is applied to. Options: DFW (applies to all VMs protected by DFW), specific Groups (apply only to VMs in selected groups), specific Segments, or logical ports. For performance, narrow the 'Applied To' scope — a rule that only needs to protect the PCI zone should have 'Applied To = PCI-Group' rather than DFW (all workloads). Narrowing scope reduces the rule evaluation overhead on hosts that have no PCI VMs. This is especially important for large DFW rule bases — unnecessary rule evaluation on irrelevant hosts wastes CPU cycles.

Q 166

What is Security Policy Automation with Tags?

Answer

NSX-T Tags enable automated security policy application through a GitOps/IaC workflow: (1) Define DFW rules referencing groups based on tags (e.g., Group 'PCI-Web' = VMs with Tag 'Environment=PCI' AND Tag 'Tier=Web'). (2) Create Terraform/Ansible scripts that deploy VMs and apply appropriate NSX-T tags. (3) When a new web VM is deployed in the PCI environment with correct tags, it is automatically included in the PCI-Web group and all DFW rules for that group are instantly applied. (4) When the VM is decommissioned, tags are removed and firewall rules are automatically unapplied. This zero-touch security automation eliminates manual firewall rule changes during VM lifecycle events.

Q 167

What is NSX-T configuration drift detection?

Answer

Configuration drift in NSX-T occurs when the actual running state of transport nodes diverges from the desired configuration in NSX Manager (e.g., a VIB is accidentally removed from an ESXi host, a manual command was run on an Edge node changing configuration). NSX-T detects drift via: Realization State checks (Policy API realization verification), Host Transport Node configuration state (should be 'Stable'), and periodic reconciliation. When drift is detected, NSX Manager shows an 'Error' or 'Partial Success' realization state for affected objects. Resolve drift: for hosts, use 'Resolve Configuration' action in NSX Manager; for manual Edge CLI changes, restore via NSX Manager API push.

Q 168

What is DFW rule logging best practice?

Answer

DFW rule logging best practices: Enable logging only on high-value rules (do not log all rules – generates massive log volume that overwhelms syslog servers and obscures important events). Always log: Default Drop rule (every blocked connection is logged – critical for security investigation), Inter-environment rules (production to DMZ, PCI zone access), privileged access rules (admin RDP/SSH), and any rule recently changed (during verification period). Log format: includes rule name, rule ID, VM name, source/destination IP, port, action. Store DFW logs in a SIEM with adequate retention (1 year for PCI-DSS). Review DFW drop logs weekly to identify misconfigurations or attack attempts.

Q 169

What is NSX-T Global Manager Federation use case?

Answer

NSX Federation Global Manager use cases: (1) Active-Active multi-site deployment: both sites run VMs; NSX Federation ensures consistent firewall policies across sites and allows workload migration between sites with IP address retention. (2) Active-Standby DR: primary site runs production; DR site is standby; NSX Federation pre-configures the DR site network so VMs can start immediately at DR site after failover. (3) Multi-datacenter policy consistency: ensure the same DFW micro-segmentation rules apply to the same application running in different data centers. (4) Centralized compliance: one security team manages consistent NSX policies across all sites from Global Manager, while local teams manage site-specific resources via Local Managers.

Q 170

What is NSX-T DFW time-based rules?

Answer

NSX-T DFW supports time-based rules (scheduled rules) that are automatically enabled or disabled at specific times. Use cases: allow developer access to production databases only during business hours (9 AM - 6 PM weekdays), restrict Internet access from VMs during maintenance windows, enable emergency access rules for a limited time period. Time-based rules are configured with a schedule profile specifying: start time, end time, time zone, and recurrence (daily, weekly, specific dates). Outside the scheduled time, the rule is automatically disabled. Time-based rules reduce the risk of human error from forgetting to re-apply or remove temporary firewall changes.

Q 171

What is NSX-T Flexible Segmentation?

Answer

NSX-T Flexible Segmentation allows mixing of workloads with different trust levels on the same physical VLAN or overlay segment, while still enforcing micro-segmentation policies between them. Instead of physically separating workloads by VLAN (coarse segmentation), DFW allows any-to-any L2 communication to be permitted or denied at the vNIC level. For example: Development and QA VMs can share the same segment while DFW prevents Dev VMs from communicating directly with QA databases. This reduces VLAN sprawl, simplifies physical network design, and delegates security enforcement from the network team to the security team via DFW policy.

Q 172

What is NSX-T DFW with Kubernetes namespaces?

Answer

NSX-T DFW integrates with Kubernetes namespaces via NCP: each Kubernetes namespace is automatically mapped to an NSX-T security group. DFW policies can be applied at the namespace level — isolating pods in one namespace from pods in another. This enables multi-tenant Kubernetes security on shared infrastructure: Team A's namespace is isolated from Team B's namespace by default (Default Drop between namespaces), with explicit DFW rules allowing specific cross-namespace communication (e.g., allow monitoring namespace to access all namespaces on port 9090 for Prometheus scraping). Namespace-level isolation maps to NSX-T Applied-To scope, ensuring DFW rules only apply to pods in relevant namespaces.

Q 173

What is NSX-T policy section locking?

Answer

NSX-T DFW Policy Section locking prevents concurrent modifications to the same policy section by multiple administrators. When Admin-A starts editing a DFW section, NSX-T locks that section for Admin-A — Admin-B sees a 'Locked by Admin-A' indicator and cannot edit the same section simultaneously. This prevents configuration conflicts and race conditions where two admins overwrite each other's changes. Locks are released when the editing admin publishes or discards changes, or after a configurable timeout (if the admin's session is idle). In large teams, establish change management processes to coordinate who edits which DFW sections and when, to minimize lock contention.

Q 174

What is NSX-T DFW performance impact on ESXi?

Answer

NSX-T DFW adds CPU overhead to ESXi hosts proportional to: number of VMs per host (more VMs = more DFW filter instances), number of DFW rules (each packet is evaluated against all applicable rules in sequence), complexity of rules (L7 App-ID rules are more CPU-intensive than simple L3 rules), and packet rate per VM. Typical overhead: 5-15% additional ESXi CPU for moderate DFW deployments. To minimize DFW CPU impact: reduce rule count via grouping (one group-based rule replaces hundreds of IP-based rules), narrow 'Applied To' scope (rules only evaluated on relevant hosts), limit L7 App-ID rules to necessary traffic (use L4 rules for most traffic), and ensure ESXi hosts have sufficient CPU headroom (30-40% reserved for infrastructure overhead including DFW).

Q 175

What is NSX-T DFW connection table management?

Answer

NSX-T DFW connection tracking table (state table) management: each ESXi host maintains a per-VM DFW connection state table. The connection table tracks active TCP/UDP/ICMP flows for stateful enforcement. Table limits: configurable per-VM connection table size (default: 128K connections per VM, max: varies by ESXi host memory). When the connection table fills up, new connections are dropped — this can happen for VMs with very high connection rates (network scanners, load balancers, busy web servers). Monitor: `vsipioctl getStats -f | grep table` (shows connection table utilization). Increase table size for high-connection VMs via NSX Manager advanced settings. Connection table is cleared when a VM is powered off or vMotioned.

NSX-T Load Balancer

22 Q&A;

Q 176

What is NSX-T Load Balancer?

Answer

NSX-T includes a software-defined load balancer that runs on NSX Edge nodes (on the Service Router). It supports both Layer 4 (TCP/UDP) and Layer 7 (HTTP/HTTPS) load balancing. NSX-T LB can replace dedicated hardware load balancers for VM and container workloads. Key features: Virtual Servers (VIPs), Server Pools with health monitors, SSL offloading (L7), persistence (source IP, HTTP cookie), application rules (for HTTP manipulation), and integration with Kubernetes Services (via NCP). LB is licensed separately (NSX Advanced Load Balancer / AVI is the enterprise LB offering).

Q 177

What is a Virtual Server in NSX-T LB?

Answer

A Virtual Server (VS) is the front-end of the load balancer — it has a VIP (Virtual IP address) and port that clients connect to. The VS receives incoming connections, applies the configured load balancing algorithm and persistence, and distributes traffic to the backend servers in the attached Pool. A VS can be L4 (TCP/UDP passthrough with basic load balancing) or L7 (HTTP/HTTPS with application-layer features like URL rewriting, host header routing, SSL offload). Each VS references a Server Pool and an Application Profile.

Q 178

What is a Server Pool in NSX-T LB?

Answer

A Server Pool is a group of backend servers (VMs or containers) that the LB distributes traffic to. Pool configuration includes: Pool Members (IP addresses and ports of backend servers), Load Balancing Algorithm (Round Robin, Weighted Round Robin, Least Connections, IP Hash), Health Monitor (Active: LB sends periodic health checks to servers; Passive: LB monitors real traffic for failures), SNAT mode (transparent or SNAT for source IP preservation). When a pool member fails its health check, the LB automatically removes it from rotation until it recovers.

Q 179

What load balancing algorithms does NSX-T support?

Answer

NSX-T LB supports: Round Robin (requests distributed sequentially across all pool members – default), Weighted Round Robin (servers receive traffic proportional to configured weight – useful when servers have different capacities), Least Connections (new requests sent to the server with fewest active connections – good for long-lived sessions), Weighted Least Connections (least connections + weight factor), IP Hash (same source IP always goes to the same server – provides session persistence without cookies, but uneven distribution if many clients share a NAT IP). For most HTTP applications, Round Robin or Least Connections is recommended.

Q 180

What is SSL Offloading in NSX-T LB?

Answer

SSL Offloading (SSL Termination) on NSX-T LB means the LB terminates the HTTPS connection from the client (decrypts SSL/TLS), performs load balancing logic at Layer 7, and then sends requests to backend servers over either plain HTTP or re-encrypted HTTPS (SSL passthrough/end-to-end SSL). Benefits: SSL/TLS decryption offloaded from backend servers (reducing their CPU load), the LB can inspect HTTP headers and cookies for L7 features (persistence, URL routing). Configure SSL offload by importing the SSL certificate into NSX Manager and attaching it to a Virtual Server's Client SSL Profile.

Q 181

What is persistence in NSX-T LB?

Answer

Persistence (also called stickiness or affinity) ensures that subsequent requests from the same client are always sent to the same backend server. NSX-T LB supports: Source IP Persistence (based on client IP – all requests from the same IP go to the same server), Cookie Persistence (LB inserts an HTTP cookie into the response; subsequent requests with that cookie are routed to the same server – more reliable than IP-based for clients behind NAT), and Application Cookie Persistence (uses an existing application session cookie for routing). Persistence timeout determines how long the binding is maintained after the last request.

Q 182

What is Active vs Passive health monitoring in NSX-T LB?

Answer

Active Health Monitor: the LB periodically sends probe requests (TCP, HTTP, HTTPS, ICMP, UDP) to each pool member to check its health. If a member fails a configurable number of probes, it is marked as 'Down' and removed from rotation until it recovers. Configurable: interval (how often to probe), timeout (how long to wait for response), rise count (how many successful probes to mark as Up), fall count (how many failures to mark as Down). Passive Health Monitor: monitors real traffic for errors — if a certain percentage of requests fail (connection refused, HTTP 5xx), the member is marked Down. Both can be used together.

Q 183

What is NSX Advanced Load Balancer (AVI)?

Answer

NSX Advanced Load Balancer (AVI Networks, acquired by VMware) is an enterprise-grade ADC (Application Delivery Controller) that provides more advanced load balancing than the built-in NSX-T LB. AVI features: real-time analytics and application visibility, automatic scaling of LB instances, Web Application Firewall (WAF), bot detection, advanced health checking, Kubernetes Ingress controller, cloud autoscaling integration. AVI replaces the NSX-T native LB for enterprise production workloads and is the recommended LB solution for NSX-T deployments requiring advanced features. NSX Advanced Load Balancer is now called 'VMware Avi Load Balancer'.

Q 184

How does NSX-T LB integrate with Kubernetes?

Answer

NSX-T LB integrates with Kubernetes via NCP (NSX Container Plugin). When a Kubernetes Service of type LoadBalancer is created, NCP automatically provisions an NSX-T Virtual Server (VIP) for that service and creates a Server Pool with the Kubernetes pod endpoints as pool members. As pods scale up/down, NCP updates the pool members automatically. The VIP is advertised externally via the T0 BGP configuration so external clients can reach Kubernetes services. This provides native cloud-like LoadBalancer service experience in on-premises Kubernetes clusters running on vSphere with NSX-T.

Q 185

What is the LB scaling model in NSX-T?

Answer

NSX-T LB scales by adding more LB service instances (each running on an Edge node SR). Each VS is assigned to a specific Edge SR instance. To scale LB capacity: add more Edge nodes to the Edge cluster, create additional VS instances (each VS can handle a defined number of connections per Edge node size – Small: 8K, Medium: 16K, Large: 90K, XL: 200K concurrent connections). For very high throughput, use multiple Large or XL Edge nodes with separate VS instances distributing load. AVI Networks (NSX ALB) provides more elastic scaling – LB instances automatically scale with demand.

Q 186

What is NSX-T LB vs NSX Advanced Load Balancer (AVI)?

Answer

NSX-T Native LB: built-in, runs on Edge nodes (SR), suitable for basic L4/L7 load balancing, simpler to configure, included in NSX-T licensing. Limitations: scaling requires adding Edge nodes, limited analytics, no WAF. NSX Advanced Load Balancer (AVI / Avi Networks): dedicated ADC software, deployed as Service Engines (SEs) on ESXi, elastic auto-scaling, real-time analytics, WAF (Web Application Firewall), bot detection, Kubernetes ingress, centralized AVI Controller. AVI is the enterprise recommendation for production workloads needing advanced features. AVI is licensed separately. For simple internal load balancing of 2-3 applications, NSX-T native LB is sufficient.

Q 187

How does NSX-T LB SNAT work?

Answer

When NSX-T LB uses SNAT (Source NAT) mode, the load balancer replaces the client's source IP with the LB's IP before forwarding the request to the backend server. The backend server sees the LB's IP as the source – not the actual client IP. This simplifies routing (backend servers only need to route back to the LB, not to all client subnets). Limitation: application loses visibility into real client IP (logging shows LB IP instead of client IP). To preserve client IP: use Transparent mode (no SNAT – backend servers must route responses directly to clients, requiring the LB to be in the default gateway path), or use X-Forwarded-For HTTP header insertion (L7 LB only – LB adds client IP in X-Forwarded-For header).

Q 188

What is Connection Mirroring in NSX-T LB?

Answer

Connection Mirroring (Traffic Mirroring) in NSX-T LB allows a copy of all client traffic to a Virtual Server to be sent to a designated mirror pool – a secondary set of servers that receive the same traffic but do not send responses back to clients. Use cases: security inspection (send a copy of all HTTP traffic to a WAF/IDS), testing (mirror production traffic to a staging environment for performance testing without real users), or passive monitoring. The primary pool handles actual client requests normally. The mirror pool receives a best-effort copy – mirrored traffic may be dropped under high load without affecting the primary traffic flow.

Q 189

What is NSX-T LB health monitor types?

Answer

NSX-T LB supports multiple health monitor types: TCP (connect to backend port – checks port is open and accepting connections), HTTP (send an HTTP GET request – checks for specific HTTP response code and optionally response body), HTTPS (same as HTTP but over SSL – verifies the application HTTPS endpoint), ICMP (ping the backend server – basic reachability check, not application-layer health), UDP (send UDP probe – limited use, as UDP is connectionless), and custom (scripted health check for complex application health verification). For production web applications, use HTTP/HTTPS health monitors with a specific health check URL (e.g., /healthz endpoint) that verifies application layer health, not just TCP connectivity.

Q 190

What is NSX-T Segment admin state?

Answer

NSX-T Segments have an Admin State that controls whether traffic is allowed on the segment: UP (default – traffic flows normally), DOWN (segment is administratively disabled – all ports are blocked, no traffic flows). Setting a segment to Down is useful for: emergency isolation of a compromised application segment, decommissioning a segment before deletion, or testing isolation scenarios. Individual logical ports also have admin state (Up/Down) for per-VM isolation without affecting the entire segment. Change segment admin state: NSX Manager > Networking > Segments > [Segment] > Edit > Admin State.

Q 191

What is the NSX-T LB access log?

Answer

NSX-T LB access log records HTTP/HTTPS requests processed by Layer 7 Virtual Servers. Log entries include: client IP, request timestamp, HTTP method and URL, HTTP response code, bytes sent, backend server IP:port, request processing time, and session persistence information. Access logs are critical for: application troubleshooting (identifying failed requests, slow responses), security forensics (identifying suspicious access patterns, DDoS attacks), and capacity planning (request rate, response time trends). LB access logs are forwarded via syslog to the same log management system as DFW logs. Configure access logging per Virtual Server in the Application Profile settings.

Q 192

What is NSX-T Kubernetes NetworkPolicy?

Answer

Kubernetes NetworkPolicy is a native Kubernetes object for defining pod-to-pod communication rules. NSX-T (via NCP) translates Kubernetes NetworkPolicy resources into NSX-T DFW rules automatically. When a developer creates a NetworkPolicy in Kubernetes (e.g., 'allow frontend pods to communicate with backend pods on port 8080'), NCP reads the NetworkPolicy and creates equivalent NSX-T DFW rules that enforce the policy at the hypervisor vNIC level. This provides: stronger enforcement than Kubernetes-native NetworkPolicy (which relies on kube-proxy, bypassable), visibility in NSX Intelligence, and consistent policy management alongside VM security policies in NSX Manager.

Q 193

What is NSX-T LB connection rate limiting?

Answer

NSX-T LB supports connection rate limiting per Virtual Server to protect backend servers from connection floods (DDoS protection). Configure: Max Concurrent Connections (maximum total simultaneous connections the VS will accept), Max New Connections per Second (rate at which new connections are accepted). When limits are exceeded, new connections are rejected (TCP RST or HTTP 503). Connection rate limiting is configured in the Virtual Server settings. For applications experiencing occasional traffic spikes (flash sales, batch job starts), set limits high enough to handle peak load but low enough to prevent backend overload. Monitor LB statistics to tune limits appropriately.

Q 194

What is NSX-T LB X-Forwarded-For header?

Answer

The X-Forwarded-For (XFF) HTTP header is inserted by NSX-T Layer 7 LB into forwarded HTTP requests to carry the original client IP address to backend servers. Without XFF, backend servers see the LB's IP as the source of all requests — losing client IP information critical for logging, analytics, geolocation, and security. With XFF enabled (in the Application Profile): LB adds 'X-Forwarded-For: ' header to each HTTP request forwarded to the backend. Backend applications must be configured to log and use the XFF header instead of the TCP source IP. XFF can be chained (if multiple proxies/LBs are in the path, each appends its IP).

Q 195

What is NSX-T LB troubleshooting methodology?

Answer

NSX-T LB troubleshooting steps: (1) Check VS status — NSX Manager > Networking > Load Balancing > Virtual Servers > [VS] — should be 'UP'. (2) Check Pool status — all pool members should be 'UP' in health monitor. If a member is 'DOWN': verify backend server is running, service is listening on configured port, LB can reach server (check routing, DFW rules between Edge LB network and backend server network). (3) Check Edge node running the LB SR — SSH to Edge, enter T1 LB VRF context, get lb-statistics. (4) Verify NAT — if using SNAT, verify backend server routes responses to LB IP. (5) Test from client to VIP, and separately from Edge to backend server to isolate client-side vs server-side issues. (6) Check LB access logs for error details.

Q 196

What is NSX-T LB active-active vs active-standby pool members?

Answer

NSX-T LB server pools support active/standby member configuration: Active members receive traffic during normal operation. Standby (Backup) members only receive traffic when all active members are down — used as last-resort failover. This is different from load distribution algorithms (Round Robin distributes across ALL active members). Example: a pool with 3 active members and 1 standby. Under normal operation, traffic distributes across the 3 active members. If all 3 fail health checks, the standby member starts receiving traffic. Configure member priority (1-65535; higher priority = active, lower = standby) in the pool member settings. Useful for disaster recovery scenarios where a standby server is in a different site.

Q 197

What is NSX-T LB for Kubernetes Ingress?

Answer

NSX-T LB integrates with Kubernetes Ingress via NCP (NSX Container Plugin). When a Kubernetes Ingress resource is created: NCP provisions an NSX-T L7 Virtual Server (HTTP/HTTPS) as the Ingress endpoint. NCP creates Server Pools for each Ingress backend service. Path-based routing (Ingress rules) is implemented as NSX-T LB application rules (HTTP header matching, URL path routing). SSL certificates for HTTPS Ingress are stored in Kubernetes Secrets and automatically imported to NSX-T for SSL offload. The Ingress VIP is advertised externally via the T0 BGP configuration. This provides a native Kubernetes Ingress experience backed by the full power of NSX-T LB — without needing a separate Ingress controller like nginx.

NSX-T VPN — IPsec & L2 VPN

21 Q&A;

Q 198

What VPN types does NSX-T support?

Answer

NSX-T supports two VPN types, both running on NSX Edge nodes (T0/T1 SR): IPsec VPN (Site-to-Site VPN) — encrypts IP traffic between two NSX-T sites or between NSX-T and a remote VPN device (Cisco, Palo Alto, FortiGate, AWS VPN Gateway). Supports both Route-Based (Routed) and Policy-Based (Legacy) VPN modes. L2 VPN (Layer 2 Extension) — extends Layer 2 segments between two NSX-T sites over an encrypted tunnel, allowing VMs to maintain their IP addresses when migrated between sites (required for workload mobility without IP re-addressing).

Q 199

What is Route-Based IPsec VPN in NSX-T?

Answer

Route-Based VPN creates a virtual tunnel interface (VTI) and uses dynamic routing (BGP) or static routes to send traffic through the tunnel. Traffic for subnets reachable via the tunnel interface is automatically encrypted and sent through the IPsec tunnel. Advantages: supports dynamic routing, more flexible than policy-based, compatible with most modern VPN devices, supports redundant tunnels with failover. Configuration: create IPsec Service on T0 or T1, define IKE Profile (IKE version, encryption, authentication), IPsec Profile (ESP encryption, PFS), create Local Endpoint and Peer, create Route-Based Session.

Q 200

What is Policy-Based IPsec VPN in NSX-T?

Answer

Policy-Based VPN creates specific traffic selectors (local subnet + remote subnet combinations) that define which traffic is encrypted and sent through the VPN. A separate SA (Security Association) is created for each traffic selector pair. Simpler for fixed, small-scale site-to-site VPN requirements. Compatible with older VPN devices that do not support route-based VPN. Limitation: does not support dynamic routing, traffic selector changes require VPN reconfiguration, and overlapping subnets cause issues. Used when the remote end (e.g., a customer's legacy firewall) only supports policy-based VPN.

Q 201

What IKE versions does NSX-T VPN support?

Answer

NSX-T VPN supports both IKEv1 and IKEv2: IKEv1 (older standard): supported for compatibility with legacy VPN devices. Uses two phases – Phase 1 (ISAKMP SA establishment) and Phase 2 (IPsec SA establishment). Supports Main Mode and Aggressive Mode. IKEv2 (recommended): more efficient (combines Phase 1/2 into fewer messages), supports multi-homing, asymmetric authentication (different auth methods for each side), and improved failover. IKEv2 is strongly recommended for new deployments. NSX-T creates IKE Profiles that specify the IKE version, DH group (2048-bit minimum recommended), encryption (AES-256-GCM recommended), and integrity algorithm.

Q 202

What is Dead Peer Detection (DPD) in NSX-T VPN?

Answer

DPD is a mechanism in IKEv1/IKEv2 that detects when the remote VPN peer is no longer reachable (peer has crashed, link is down). Periodically sends DPD probes through the established IKE SA. If the peer does not respond within the configured DPD timeout, the VPN tunnel is torn down and re-established (or failover to a secondary tunnel is triggered). DPD is configured in the IKE Profile with: DPD probe interval (how often to send probes), DPD probe count (how many missed probes before declaring the peer dead). DPD is critical for detecting tunnel failures faster than waiting for routing protocol convergence.

Q 203

What is NSX-T L2 VPN and when is it used?

Answer

L2 VPN (Layer 2 Extension) creates an encrypted tunnel between two NSX-T sites (or NSX-T to a standalone NSX-T Edge) that extends a Layer 2 segment across sites. Use cases: Workload migration without IP re-addressing – a VM can be vMotioned from Site A to Site B and keep its original IP address because its subnet is stretched via L2 VPN. Disaster recovery – replicated VMs at the DR site have the same IP addresses as production. Hybrid cloud – stretch on-premises VLANs to VMware Cloud on AWS. L2 VPN uses NSX-T Edge nodes on both ends (client Edge and server Edge) connected by an IPsec-encrypted tunnel.

Q 204

What encryption algorithms does NSX-T VPN support?

Answer

NSX-T VPN supports: IKE (Phase 1) Encryption: AES-128, AES-256, AES-256-GCM (recommended), AES-128-GCM. DH Groups: Group 2 (1024-bit – not recommended), Group 5 (1536-bit), Group 14 (2048-bit – minimum recommended), Group 15/16/19/20/21 (3072-4096-bit elliptic curve). IPsec (Phase 2) Encryption: AES-GCM-128, AES-GCM-256 (recommended – provides both encryption and integrity), AES-128-CBC, AES-256-CBC. Integrity (when not using GCM): SHA-256, SHA-384, SHA-512. For PCI-DSS or FIPS compliance, use AES-256-GCM with DH Group 14+ and SHA-256+.

Q 205

How do you monitor VPN tunnel status in NSX-T?

Answer

VPN tunnel monitoring options: NSX Manager UI – Networking > VPN > Status tab shows IKE SA and IPsec SA status (Up/Down), last error, bytes in/out. NSX Manager API – GET /policy/api/v1/infra/tier-0s/{id}/ipsec-vpn-services/{id}/sessions/{id}/statistics for tunnel statistics. Edge CLI – get vpn ipsec sa (show IPsec SAs), get vpn ipsec status (overall VPN status), get vpn ike sessions (IKE session status). NSX-T Alarms – configure alarms for VPN session Down events to trigger email/SNMP notifications. Syslog – NSX-T logs VPN events (tunnel Up/Down, rekey events) to syslog for SIEM integration.

Q 206

What is Perfect Forward Secrecy (PFS) in NSX-T VPN?

Answer

PFS (Perfect Forward Secrecy) is a VPN security feature that generates a new Diffie-Hellman key pair for each IPsec Phase 2 SA (IPsec SA rekey), independent of the Phase 1 (IKE SA) key material. This means that if the long-term Phase 1 keys are compromised, past encrypted sessions cannot be decrypted because each session used a unique ephemeral Phase 2 key. NSX-T supports enabling PFS with DH Groups 2, 5, 14, 15, 16, 19, 20, 21. PFS is enabled in the IPsec Profile. DH Group 14 (2048-bit) or higher is recommended for modern deployments. PFS adds a small computational overhead but is strongly recommended for sensitive data.

Q 207

What is VPN high availability in NSX-T?

Answer

NSX-T VPN high availability approaches: (1) T0 Active-Standby with VPN: one Edge SR handles VPN; if it fails, standby takes over (VPN tunnels must re-establish after failover – brief interruption). (2) Route-Based VPN with redundant tunnels: configure two VPN sessions (primary and secondary) to the same remote peer via different Edge uplinks or different physical paths; use BGP with local preference to prefer primary tunnel; BGP automatically fails over to secondary if primary goes down. (3) BFD-monitored VPN: configure BFD on BGP sessions over VPN tunnels for fast failure detection and quick failover.

Q 208

What is NSX-T VPN failover configuration?

Answer

NSX-T VPN failover for high availability: (1) Create two IPsec sessions to the same remote peer – primary (via uplink-1) and secondary (via uplink-2 or different Edge node). (2) Use Route-Based VPN with BGP: advertise BGP routes over both tunnels. Set higher local preference on primary tunnel routes. When primary fails, BGP converges to secondary tunnel. (3) Configure DPD on both sessions for fast peer failure detection. (4) For T0 active-active with two Edge nodes: each Edge has its own IPsec session to the remote peer; physical routers use ECMP to distribute traffic. Both sessions run simultaneously, providing both redundancy and load balancing for VPN traffic.

Q 209

What is NSX-T L2 VPN client and server mode?

Answer

NSX-T L2 VPN operates in client-server mode: L2 VPN Server: runs on an NSX-T Edge node (T0 or T1 SR). It is the central endpoint that accepts L2 VPN connections from remote sites. Configured with a server certificate, allowed subnets, and a local tunnel endpoint IP. L2 VPN Client: connects to the L2 VPN Server. Can be another NSX-T Edge node (for NSX-T to NSX-T site-to-site), or a standalone NSX-T Edge VM deployed at a remote location without full NSX-T deployment (Edge node in standalone mode). The tunnel uses IPsec encryption for security. Each VPN session extends one or more NSX-T segments from the server site to the client site.

Q 210

What are common IPsec VPN troubleshooting steps in NSX-T?

Answer

IPsec VPN troubleshooting steps: (1) Check IKE SA status – Edge CLI: get vpn ike sessions (should show Established). If not, check: firewall between Edge uplink and remote peer (allow UDP 500/4500), IKE profile mismatch (version, encryption, DH group must match both sides), pre-shared key mismatch, peer IP misconfiguration. (2) Check IPsec SA status – get vpn ipsec sa (should show Active). If IKE is up but IPsec down, check IPsec profile mismatch (ESP encryption, PFS settings). (3) Check routing – verify routes for remote subnets appear in T0 routing table (get route in T0 VRF context). (4) Check DPD – ensure DPD probe interval matches between peers. (5) Packet capture on Edge uplink interface for detailed debugging.

Q 211

What is NSX-T Logical Port statistics?

Answer

NSX-T Logical Port statistics provide per-VM vNIC traffic metrics: Bytes/Packets Transmitted (TX) and Received (RX), Drop counters (DFW drops, hardware drops), Broadcast/Multicast packet counts. Access via: NSX Manager > Networking > Segments > [Segment] > Ports > [Port] > Statistics tab. Via API: GET /api/v1/logical-ports/{id}/statistics. These statistics are valuable for: capacity planning (identifying high-bandwidth VMs), security forensics (unusual traffic volume from a specific VM), and troubleshooting (identifying packet drops on a specific port). Statistics are sampled every 60 seconds and stored in NSX Manager for trending.

Q 212

What is TCP Multiplexing in NSX-T LB?

Answer

TCP multiplexing (TCP connection pooling) is an NSX-T LB performance optimization where the LB maintains persistent TCP connections to backend servers and reuses them for multiple client requests. Without multiplexing: each client request requires a new TCP three-way handshake to the backend server (adding latency). With multiplexing: the LB keeps a pool of pre-established connections to backend servers; client requests are forwarded over existing connections immediately. Benefits: reduced backend server connection overhead, lower latency for individual requests, better handling of high request rates. TCP multiplexing is configured in the Application Profile attached to a Virtual Server.

Q 213

What is NSX-T and Ansible automation?

Answer

The community.vmware.nsxt Ansible Collection provides modules for automating NSX-T: nsxt_policy_segment (create/manage segments), nsxt_policy_tier0 and nsxt_policy_tier1 (gateway management), nsxt_policy_security_policy and nsxt_policy_group (firewall automation), nsxt_policy_ip_pool (IP pool management), nsxt_transport_node (host preparation), nsxt_policy_lb_virtual_server (LB automation). Ansible playbooks can be used for: automated environment provisioning (create a full network topology for a new application in minutes), configuration drift correction (ensure NSX-T config matches desired state), compliance enforcement (run playbooks that verify and correct security policy compliance). NSX-T Ansible modules use the REST API under the hood.

Q 214

What is NSX-T VPN IKE SA rekey?

Answer

IKE SA rekeying is the process of establishing a new IKE Security Association before the current one expires, ensuring continuous VPN tunnel operation without disruption. NSX-T automatically rekeys IKE SAs based on the lifetime configured in the IKE Profile (default: 86400 seconds / 24 hours). Rekeying begins before expiry (at approximately 90% of lifetime). Similarly, IPsec SAs are rekeyed based on the IPsec Profile lifetime (default: 3600 seconds / 1 hour) or byte count (default: 100 GB). Properly functioning rekeying means VPN tunnels run indefinitely without manual intervention. If rekeying fails (mismatched configuration or network issue), the tunnel drops at SA expiry – monitor DPD and SA expiry alarms.

Q 215

What is NSX-T L2 VPN traffic optimization?

Answer

NSX-T L2 VPN traffic optimization considerations: (1) Only stretch segments that truly need L2 extension – routing (T0/T1) is preferred over L2 stretching for performance and stability. (2) Use ARP suppression on stretched segments to minimize broadcast traffic across the WAN link. (3) Monitor L2 VPN tunnel bandwidth utilization – L2 extension carries all broadcast and multicast traffic, which can consume significant bandwidth. (4) Consider setting DHCP relay on stretched segments to keep DHCP local at each site rather than crossing the L2 VPN. (5) Implement QoS on the WAN uplinks to prioritize L2 VPN traffic over less critical traffic. (6) Plan migration path to routing – L2 VPN should be a transitional tool, not a permanent architecture.

Q 216

What is NSX-T VPN and overlapping IP addresses?

Answer

Overlapping IP addresses between local NSX-T networks and remote VPN-connected networks cause routing confusion — the router cannot distinguish between local and remote destinations for the same IP range. Handling: (1) Best solution: redesign IP addressing to eliminate overlap (use unique RFC 1918 ranges for each site). (2) When redesign is not possible: use Policy-Based VPN with explicit traffic selectors (specify exact local and remote IP ranges to tunnel) — this forces matching traffic into the VPN regardless of routing ambiguity. (3) Use NAT on the VPN: translate one side's addresses to non-overlapping ranges before the VPN traffic enters the tunnel (NAT before VPN — supported in NSX-T with careful rule ordering). (4) Document all IP addressing before VPN configuration to proactively identify overlaps.

Q 217

What is NSX-T VPN certificate-based authentication?

Answer

NSX-T IPsec VPN supports certificate-based IKE authentication as an alternative to pre-shared keys (PSK). Certificate authentication is more secure (no shared secret to compromise) and scales better (each remote site has its own certificate rather than sharing a secret). Configuration: import CA certificate and site certificates in NSX Manager (System > Certificates). Create IKE Profile with Authentication Mode = RSA (uses RSA certificates). Configure Local ID and Peer ID using Distinguished Names (DN) from the certificates. Certificate-based VPN requires PKI infrastructure (CA server) on both ends. NSX-T's VMCA can issue certificates, or use an enterprise CA. Certificate expiry management is critical — expired certificates cause VPN tunnel failures.

Q 218

What is NSX-T VPN monitoring via API?

Answer

NSX-T VPN monitoring via REST API: GET /policy/api/v1/infra/tier-0s/{id}/ipsec-vpn-services/{service-id}/sessions/{session-id}/statistics — returns IKE SA status (state, rekeying info, SA establishment time), IPsec SA statistics (encrypted/decrypted bytes, packets, errors), and DPD status. For all sessions in bulk: GET /policy/api/v1/infra/tier-0s/{id}/ipsec-vpn-services/{service-id}/sessions — lists all sessions with status. Integrate these API calls into monitoring scripts (Python, PowerShell) that poll every 60 seconds and alert when session state changes from 'UP' to 'DOWN'. Include VPN monitoring in your NOC dashboard — VPN tunnel failures directly impact site-to-site connectivity and should trigger immediate response.

NSX-T Security – Advanced Features

25 Q&A;

Q 219

What is NSX-T Intrusion Detection System (IDS/IPS)?

Answer

NSX-T Distributed IDS/IPS (part of NSX Advanced Threat Prevention) is a signature-based intrusion detection and prevention system running inline in the NSX distributed dataplane – on each ESXi host at the VM's vNIC level. Unlike traditional IDS/IPS appliances that require traffic to be hairpinned through them, NSX Distributed IDS inspects East-West traffic directly on the host where VMs are running. Signatures from VMware Threat Intelligence are updated automatically. IDS/IPS can detect and optionally block known attack patterns, CVE exploits, and malicious protocols in real time without impacting workload performance significantly.

Q 220

What is NSX-T Network Detection and Response (NDR)?

Answer

NSX Network Detection and Response (NDR) – part of NSX Advanced Threat Prevention – uses AI/ML-based behavioral analytics to detect advanced persistent threats (APTs), lateral movement, and command-and-control (C2) communication that signature-based IDS might miss. NDR correlates network flows from DFW, IDS events, and NSX Intelligence data to build behavioral baselines and detect anomalies. When a threat campaign is detected, NSX NDR provides an attack campaign view showing the attack chain, affected workloads, and recommended containment actions.

Q 221

What is NSX-T URL Filtering?

Answer

URL Filtering in NSX-T is a Layer 7 DFW feature that allows or blocks outbound web traffic based on URL categories (Social Media, Malware, Gambling, Streaming, etc.) and specific URLs/domains. It is configured via Context Profiles in DFW rules. URL categories are maintained by Brightcloud/VMware Threat Intelligence and updated automatically. Example use case: Block all VMs in the PCI zone from accessing any URLs not in the 'Payment Processing' whitelist. URL Filtering requires the NSX Distributed Firewall with Advanced Threat Prevention license and the NSX Intelligence appliance.

Q 222

What is NSX-T DNS Security?

Answer

NSX-T DNS Security (part of NSX Advanced Threat Prevention) monitors DNS traffic from VMs and detects/blocks DNS-based attacks: DNS tunneling (data exfiltration via DNS), domain generation algorithm (DGA) malware beaconing, DNS rebinding attacks, and queries for known malicious domains (blocked by threat intelligence feed). DNS Security works by distributing DNS inspection to each ESXi host's dataplane — inspecting DNS queries/responses at the VM level without requiring DNS traffic to traverse a centralized inspection appliance. It integrates with NSX Distributed IDS for correlated threat detection.

Q 223

What is NSX-T Malware Prevention?

Answer

NSX-T Distributed Malware Prevention (part of NSX Advanced Threat Prevention) provides inline file inspection for files transferred over the network (HTTP, FTP, SMB). When a file is detected in a network flow, NSX-T extracts the file and submits it to the VMware Advanced Threat Prevention cloud (or on-premises sandbox) for malware analysis. If the file is determined to be malicious, the DFW can block the transfer and quarantine the affected VM. This provides 'last line of defense' malware protection for East-West traffic — catching malware that has bypassed perimeter controls.

Q 224

What is NSX-T Service Accounts and how do they work?

Answer

NSX-T supports Service Accounts for non-interactive authentication from automation tools (Terraform, Ansible, CI/CD pipelines) to the NSX-T API. Service accounts use either: Local Service Accounts (created in NSX Manager with a specific role — Operator, Read-Only, etc.), Principal Identity accounts (certificate-based authentication — no password, used by orchestration systems like vRA), or AD/LDAP accounts via vIDM integration (VMware Identity Manager). Service accounts should follow least-privilege — create separate accounts for each automation tool with only the permissions needed for that tool's specific operations.

Q 225

What is NSX-T Role-Based Access Control (RBAC)?

Answer

NSX-T RBAC controls who can access NSX Manager and what they can do. Built-in roles: Enterprise Admin (full access), Network Admin (networking config, no security), Security Admin (DFW, IDS, security groups — no networking), Auditor (read-only), LB Admin (load balancer only), VPN Admin (VPN only), and more. Custom roles can be created in NSX-T 3.1+. RBAC integrates with vIDM (VMware Identity Manager) for SSO with Active Directory. For multi-tenant environments, NSX-T Projects (NSX 4.x) provide role assignments scoped to specific project resources.

Q 226

What is NSX-T Context Profiles?

Answer

Context Profiles in NSX-T DFW extend firewall rules beyond IP/port matching to include: L7 App-IDs (identify the application regardless of port — e.g., SSL, HTTP, DNS, RDP), Domain Name (URL/FQDN-based matching), Custom Attributes (matching on TLS SNI, HTTP Host header, URL path). Example: Allow traffic from App-VMs to Internet on port 443 only if the destination domain is *.microsoft.com (blocking all other HTTPS traffic). Context Profiles require the App-ID license and increase CPU on hosts proportionally to inspection depth. Multiple App-IDs can be combined in one Context Profile for complex rules.

Q 227

What is NSX-T Gateway Firewall?

Answer

Gateway Firewall (Edge Firewall) is a stateful firewall running on NSX Edge nodes on T0/T1 gateways, inspecting North-South traffic entering or leaving the NSX overlay fabric. Unlike DFW (which is distributed on each host), Gateway Firewall is centralized on Edge nodes. It inspects traffic at the gateway level — ideal for protecting the perimeter between external networks and the overlay. Gateway Firewall rules are configured per T0 or T1 gateway interface (uplink, downlink). Supports the same rule structure as DFW (Source/Destination/Services/Action) plus Edge-specific features like geolocation-based blocking.

Q 228

What is NSX-T Micro-segmentation vs Traditional VLAN segmentation?

Answer

Traditional VLAN-based segmentation: workloads are grouped by VLAN (e.g., VLAN 100 = Web Tier, VLAN 200 = DB Tier); a physical or virtual firewall inspects inter-VLAN traffic. Problems: coarse granularity (all VMs in the same VLAN are implicitly trusted), lateral movement risk (if one web VM is compromised, it can attack other web VMs on the same VLAN), traffic must traverse a central firewall causing bottlenecks. NSX-T Micro-segmentation: DFW enforces firewall rules per VM's vNIC regardless of VLAN; rules are based on workload identity (tags/groups) not IP/VLAN; East-West traffic is inspected locally on each host; zero trust between every workload. Micro-segmentation eliminates lateral movement even within the same VLAN.

Q 229

What is Lateral Movement and how does NSX-T prevent it?

Answer

Lateral movement is a cyberattack technique where an attacker who has compromised one VM (e.g., a web server) moves to other VMs within the same network to access more valuable targets (databases, domain controllers). Traditional firewalls only inspect North-South traffic – they cannot see traffic between VMs on the same VLAN/segment. NSX-T DFW inspects all VM-to-VM traffic regardless of network location. With micro-segmentation: each VM is isolated by default (Default Drop rule), and only explicitly defined communication flows are allowed. Even if an attacker compromises a web VM, they cannot reach any other VM unless a DFW rule explicitly permits it.

Q 230

What is the NSX-T Audit Log?

Answer

NSX-T Audit Log records all configuration changes made via the UI, API, or CLI: what was changed (policy name, rule details), who made the change (username, IP), and when. Accessible via NSX Manager UI: System > Logs > Audit Logs, or via API: GET /api/v1/audit-logs. Audit logs should be forwarded to an external SIEM (Splunk, Elastic) via syslog for long-term retention and compliance (PCI-DSS requires 1 year log retention). NSX-T also supports RBAC audit events (logins, permission changes) in the audit log. Critical for security investigations and compliance audits.

Q 231

What is NSX-T Distributed IDS signature management?

Answer

NSX-T Distributed IDS uses threat signatures maintained by VMware's Threat Intelligence team. Signatures are automatically updated via VMware's cloud-based signature repository (requires internet access from NSX Manager, or an offline update package for air-gapped environments). Signature updates can be configured as automatic (recommended) or manual. Signature sets can be filtered – you can choose to enable only signatures relevant to your environment (e.g., only Windows workload signatures if you have no Linux VMs). False-positive management: tune IDS by excluding specific signatures or creating exceptions for known-safe traffic patterns that trigger IDS alerts.

Q 232

What is NSX-T Zero Trust security model?

Answer

Zero Trust with NSX-T implements the principle 'never trust, always verify' for workload communication: (1) Default Deny: change the DFW default rule to Drop – no communication is allowed unless explicitly permitted. (2) Micro-segmentation: define DFW rules allowing only required communication flows (application A to DB on port 5432 only). (3) Identity-based access: use IDFW to tie firewall rules to user identities (not just IPs). (4) Continuous verification: NSX IDS monitors allowed traffic for anomalies. (5) Least privilege: each workload can only access exactly what it needs. (6) Visibility: NSX Intelligence maps all actual flows to identify violations. Zero Trust with NSX-T ensures lateral movement is impossible even after a workload is compromised.

Q 233

What is NSX-T security tagging automation?

Answer

Security tagging automation in NSX-T enables dynamic security response: (1) Define a 'Quarantine' tag and a corresponding DFW policy: VMs with tag 'Status=Quarantine' are blocked from all communication except to IT remediation servers. (2) When a security incident is detected (by IDS, SIEM, or manual investigation), automatically apply the 'Quarantine' tag to the affected VM via NSX-T API (one API call). (3) The VM is immediately isolated – DFW rules for the Quarantine group take effect within seconds. (4) After remediation, remove the tag – the VM automatically rejoins its original security groups and regains normal connectivity. This reduces attacker dwell time from hours (manual isolation) to seconds (automated tag-based response).

Q 234

What is NSX-T Endpoint Security Framework?

Answer

NSX-T Endpoint Security Framework (ESF) provides hooks at the hypervisor level for third-party security vendor integration. Security vendors (Trend Micro, Carbon Black, Symantec) deploy Service VMs (SVMs) on each ESXi host. NSX-T ESF redirects VM activity events (network packets, file I/O) to the SVM for inspection without installing an agent in each VM. The SVM analyzes events and can instruct NSX-T to block specific traffic (via the Endpoint Protection policy in DFW). ESF is used for: agentless antivirus/anti-malware, network traffic deep inspection, and file integrity monitoring – all without touching the guest OS.

Q 235

What is NSX-T Private IP (RFC 1918) vs Routable IP for VMs?

Answer

In NSX-T overlay environments, VMs typically use private RFC 1918 IP addresses (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) on their segments. These addresses are not routable on the internet or physical corporate network directly. NSX-T NAT (on T0/T1 gateways) translates private IPs to routable IPs for North-South communication. This provides additional security (VMs are not directly reachable from external networks) and IP address conservation. Routable IPs are only needed for services that require direct external access (web servers, VPN endpoints, management). DFW rules use group-based policies that work equally for private or public IPs.

Q 236

What is the NSX-T T0 HA VIP?

Answer

In active-standby T0/T1 configurations, a HA VIP (High Availability Virtual IP) is configured on the T0/T1 uplink or downlink interface. The HA VIP is a shared IP address held by the active SR. When failover occurs, the standby SR takes over and assumes the VIP address. This allows the connected devices (physical routers for T0 uplink, or VMs for T1 downlink gateway IP) to maintain a stable IP address that does not change during failover – avoiding the need to update routes or ARP tables on external devices. HA VIP requires both SRs to be in the same L2 domain (same VLAN) for the IP to move between them.

Q 237

What is NSX-T Remote Access VPN (SSL VPN)?

Answer

NSX-T does not natively support SSL/TLS-based remote-access VPN (client-to-site). NSX-T VPN is focused on site-to-site IPsec and L2 extension. For remote access VPN (employees connecting from laptops): use VMware SD-WAN (VeloCloud) which includes SSL VPN capability, or deploy a third-party VPN gateway (Cisco AnyConnect, Palo Alto GlobalProtect) in the DMZ segment as a VM, with NSX-T DFW controlling access from the VPN-connected users to internal segments. NSX-T Gateway Firewall can enforce security policies on traffic from the VPN gateway to internal workloads, providing consistent security for remote users.

Q 238

What is NSX+ (NSX Cloud-Delivered)?

Answer

NSX+ is VMware's cloud-delivered SaaS version of NSX management (introduced in NSX 4.1). Instead of deploying NSX Manager on-premises, NSX+ provides the management plane as a SaaS service hosted by VMware. Customers connect their on-premises NSX data planes (transport nodes, Edge nodes) to the NSX+ cloud manager. Benefits: no NSX Manager VMs to maintain (patching, HA, backup handled by VMware), always on the latest version, elastic scale, and VMware-managed SLAs. NSX+ is positioned for customers wanting to reduce operational overhead. The data plane (actual packet forwarding, DFW enforcement) still runs on-premises — only the management plane is cloud-delivered.

Q 239

What is NSX-T IDS signature exclusion?

Answer

NSX-T Distributed IDS signature exclusions allow specific IDS signatures to be disabled for all workloads or for specific groups — preventing false positives from triggering alerts or blocks on legitimate traffic. Use cases: a known-safe application triggers an IDS signature (e.g., a security scanner VM triggers SQLi detection signatures), or a legacy application uses protocols that match malware signatures but is known clean. Configuration: NSX Manager > Security > Intrusion Detection > Signature Management > Exclude specific signatures (by CVE ID or signature ID). Document all exclusions with business justification — regular review of exclusions is required to avoid security gaps from outdated exclusions.

Q 240

What is NSX-T East-West vs North-South threat detection?

Answer

East-West threat detection in NSX-T: Distributed IDS/IPS (on each ESXi host) detects threats in VM-to-VM traffic directly on the host – catches lateral movement, exploitation of internal vulnerabilities, and C2 traffic between compromised VMs. This is invisible to perimeter defenses. North-South threat detection: NSX Edge Gateway Firewall with IDS/IPS detects threats in traffic entering or leaving the overlay – external attacks, data exfiltration, malware downloads. Best practice: deploy both E-W and N-S IDS for complete coverage. E-W IDS is unique to NSX-T – traditional security tools cannot inspect E-W traffic without NSX-T's distributed architecture.

Q 241

What is NSX-T FIPS compliance configuration?

Answer

Enabling FIPS mode in NSX-T: FIPS 140-2 compliance requires all cryptographic operations to use FIPS-validated modules. In NSX-T: (1) Enable FIPS mode in NSX Manager system settings (System > Configuration > FIPS Mode). (2) FIPS mode restricts available cipher suites – only FIPS-approved algorithms are enabled (AES-256-GCM, SHA-256+, RSA-2048+). (3) Non-FIPS algorithms (DES, 3DES, MD5, RC4) are disabled. (4) IPsec VPN IKE and ESP profiles must use only FIPS-approved algorithms. (5) TLS for NSX Manager API/UI uses only FIPS-approved cipher suites. Warning: enabling FIPS mode may break existing VPN tunnels if the remote peer uses non-FIPS algorithms. Plan a maintenance window and verify all IKE/IPsec profiles before enabling FIPS mode.

Q 242

What is NSX-T and PCI-DSS compliance?

Answer

NSX-T supports PCI-DSS compliance for Cardholder Data Environments (CDE): Requirement 1 (Network security controls) – DFW micro-segmentation isolates CDE from non-CDE using Default Drop + explicit Allow rules. Requirement 2 (Secure configurations) – NSX-T Security Configuration Guide hardening applied to all components. Requirement 7 (Access control) – RBAC with least-privilege for NSX-T admin access. Requirement 10 (Audit logs) – DFW flow logging forwarded to SIEM with 1-year retention. Requirement 11 (Security testing) – NSX Intelligence flow maps provide evidence of intended-only communication. NSX-T DFW is recognized by QSAs (PCI assessors) as an approved network security control for CDE segmentation when properly configured.

Q 243

What is NSX-T Gateway Firewall applied to T0?

Answer

Gateway Firewall on T0 (Tier-0 Gateway Firewall) inspects all traffic passing through the T0 gateway interface — both inbound (external to overlay) and outbound (overlay to external). Configuration: NSX Manager > Security > Gateway Firewall > Gateway Specific Rules > T0 Gateway name > add rules. Rules follow the same structure as DFW (Source, Destination, Service, Action). Key T0 Gateway Firewall use cases: block specific external IPs or subnets from entering the overlay (IP reputation blocking, geolocation blocking), enforce egress traffic control (which overlay subnets can reach the internet), and log all North-South traffic for compliance. T0 Gateway Firewall is processed by the Edge SR — traffic must traverse the Edge for inspection.

NSX-T Operations & Troubleshooting

27 Q&A;

Q 244

What is the Traceflow tool in NSX-T?

Answer

Traceflow is NSX-T's built-in packet simulation tool for troubleshooting connectivity and firewall issues without generating real traffic. You inject a synthetic packet (specifying source VM, destination IP, protocol, port) and NSX-T traces its path through the overlay, showing: which logical components (switches, routers, DFW) the packet traversed, whether it was forwarded or dropped, which DFW rule caused a drop (by rule name/ID), the TEP hops between hosts, and the final disposition (delivered to destination or dropped). Traceflow is available in NSX Manager UI (Tools > Traceflow) and via API.

Q 245

What is the Port Connection view in NSX-T?

Answer

The Port Connection view (NSX Manager > Networking > Segments > [Segment] > Ports) shows all logical ports on a segment with their status, attached VM, host, and statistics. It is useful for verifying that VMs are correctly connected to the right segment and for checking whether a port is administratively blocked by DFW or has link issues. The Effective Firewall Rules view on a port shows all DFW rules currently applied to that specific VM/port, including rules from parent groups — making it easy to verify or debug security policy for a specific workload.

Q 246

What are common causes of connectivity failures in NSX-T?

Answer

Common NSX-T connectivity failure causes: (1) TEP connectivity — MTU mismatch (Geneve overhead drops packets), TEP IP pool exhaustion, physical switch port not allowing TEP VLAN. (2) Transport zone misconfiguration — segment in wrong transport zone, host not added to transport zone. (3) DFW blocking — Default Drop rule, incorrect group membership, rule priority issue (rule not applied or wrong rule matching). (4) Routing — T1 not advertising segment routes to T0, T0 BGP session down, missing static route. (5) Edge node issue — Edge SR down, BGP peer down, uplink VLAN misconfigured. (6) Host preparation failure — NSX VIBs not installed correctly on ESXi host.

Q 247

How do you check TEP connectivity in NSX-T?

Answer

TEP connectivity verification steps: (1) NSX Manager > System > Fabric > Nodes > Host Transport Nodes – check TEP IP assigned to each host. (2) On ESXi host via SSH: `esxcfg-vmknics -l` (show VMkernel ports including TEP vmk). (3) Ping test between TEPs: on ESXi host: `vmkping -d -s 1572` (tests with Geneve-overhead-size packet; -d = don't fragment). If ping fails with large packets but succeeds with small packets, MTU is the issue. (4) Check physical switch allows TEP VLAN: `esxcli network vswitch dvs vmware vds list` (shows N-VDS uplinks). (5) NSX Manager Traceflow tool for end-to-end path verification.

Q 248

What is the NSX-T Manager API for operations?

Answer

Key NSX-T Manager API endpoints for operations: `GET /api/v1/transport-nodes` (list all transport nodes and their status), `GET /api/v1/transport-nodes/{id}/state` (transport node configuration state – Stable/Pending), `GET /api/v1/logical-ports` (list all logical ports), `GET /policy/api/v1/infra/realized-state/enforcement-points/default` (check all realized state), `GET /api/v1/alarms` (active alarms), `GET /policy/api/v1/infra/tier-0s/{id}/routing-table` (T0 routing table), `POST /api/v1/firewall/sections/{id}/rules?action=export_rules` (export DFW rules). Use the API Explorer at <https://nsx-manager/policy/api/v1/spec> for interactive documentation.

Q 249

How do you verify DFW rules are applied to a VM in NSX-T?

Answer

Verification methods: (1) NSX Manager UI – select the VM's logical port > Actions > Effective Firewall Rules – shows all DFW rules currently applied. (2) NSX Manager UI > Security > Distributed Firewall > Filter by 'Applied To' = VM name. (3) ESXi host CLI: `summarize-dvfilter | grep -A5` (shows DFW filter name for each vNIC). Then: `vsipioctl -i getrules` (shows actual rules applied). (4) ESXi host CLI: `vsipioctl stats -d` (shows per-rule packet/byte counters – useful for identifying which rule is matching traffic). (5) Traceflow with the VM as source to verify Allow/Drop behavior.

Q 250

What is NSX-T Realization State and why does it matter?

Answer

Realization State describes whether a configuration change made in NSX Manager (Policy API) has been successfully pushed to and applied on all transport nodes. When you create or modify a policy object (segment, firewall rule, route), NSX-T must propagate this change to all relevant transport nodes via the CCP. States: Success (config realized on all nodes), In Progress (still propagating), Error (failed to realize on one or more nodes – check error details). If realization fails, the configuration exists in NSX Manager but is NOT enforced on the transport nodes. Check realization state via: NSX Manager UI (object status), API: `/infra/realized-state/`.

Q 251

What is the NSX-T Central CLI?

Answer

The NSX-T Central CLI (available via NSX Manager UI: Tools > NSX CLI or via SSH to NSX Manager) allows running NSX-T diagnostic commands remotely on any transport node (ESXi host or Edge node) from the NSX Manager console without SSH-ing to each node individually. Commands: `get logical-routers` (on Edge), `get forwarding`, `get bgp neighbor`, `ping` from a specific Edge. The Central CLI is especially useful for large-scale environments where SSH access to individual hosts may be restricted. All CLI commands and their output are logged for audit purposes.

Q 252

What are NSX-T Alarms and how are they configured?

Answer

NSX-T has a built-in alarm system that generates alerts for: transport node disconnection, TEP failure, Edge node down, BGP session failure, certificate expiry, high resource utilization (CPU, memory, disk on NSX Manager), DFW capacity limits, and more. Alarms are visible in NSX Manager UI (Home > Alarms). Notifications: configure SNMP trap receivers (System > Settings > Notification) or syslog forwarding to send alarms to external monitoring systems (Nagios, SolarWinds, Splunk). Alarms have severities: Critical, High, Medium, Low, Info. Suppression rules can temporarily silence known-false-positive alarms during maintenance windows.

Q 253

What is the NSX-T upgrade process?

Answer

NSX-T upgrade process: (1) Download new NSX-T upgrade bundle from VMware Customer Connect. (2) Upload to NSX Manager (System > Upgrade). (3) Run pre-check to verify all components are upgrade-ready (no pending config issues). (4) Upgrade NSX Manager cluster first (the upgrade wizard handles rolling upgrade of all 3 manager nodes). (5) Upgrade Edge nodes next (one at a time, in maintenance mode – traffic fails over to standby/other Edge). (6) Upgrade host transport nodes last (using maintenance mode + vMotion – VMs keep running; rolling host-by-host upgrade). (7) Post-upgrade validation – verify all transport nodes are in Stable state, BGP sessions up, DFW rules realized.

Q 254

What is NSX-T Backup and Restore?

Answer

NSX-T backup configuration: NSX Manager automatically backs up its configuration (policies, objects, certificates, users) to a remote SFTP server on a configurable schedule. Configure in NSX Manager: System > Backup & Restore > Configure. Backup stores inventory, configuration, and audit logs (not VM data). Restore: deploy a fresh NSX Manager cluster (same version as backup), point to the SFTP backup location, and restore configuration. The restore does NOT restore the fabric state (transport nodes, host preparation) – those must be re-configured or will be automatically re-discovered if hosts are still configured. NSX-T backups should run at minimum daily; before major changes, take a manual backup.

Q 255

How do you check BGP status on NSX-T Tier-0?

Answer

BGP status verification: (1) NSX Manager UI – Networking > Tier-0 Gateways > [T0] > Routing > BGP – shows BGP neighbors and their state (Established/Idle/Active). (2) Edge Node CLI – SSH to Edge node, then: vrf (enter T0 SR context), then: get bgp neighbor summary (shows all BGP peers, state, up/down time, prefixes received/sent). (3) Central CLI from NSX Manager: nsxcli -h get bgp neighbor summary. (4) For route table verification: get route (shows BGP-learned routes). Common BGP issues: wrong AS number, incorrect neighbor IP, network reachability issue between Edge uplink IP and router IP, firewall blocking TCP 179.

Q 256

What is the NSX-T Operations Dashboard?

Answer

NSX-T Manager has an Operations Dashboard (Home page and System > Overview) that provides: System Health (manager nodes status, Edge node status, host transport node status), Alarms summary (count by severity), Capacity overview (number of segments, ports, firewall rules, LB instances), Recent configuration changes (last 10 operations), and quick links to common troubleshooting tools (Traceflow, Port Check, BGP status). For more advanced monitoring, NSX-T integrates with VMware Aria Operations (vRealize Operations) via the NSX-T Management Pack, providing historical performance data, capacity forecasting, and cross-product correlation.

Q 257

What is host preparation failure in NSX-T and how to fix it?

Answer

Host preparation failure means NSX-T failed to install or configure the NSX kernel modules (VIBs) on an ESXi host. Common causes: (1) ESXi version not compatible with NSX-T version – check NSX-T compatibility matrix. (2) Host in maintenance mode – cannot install VIBs on a host in maintenance mode. (3) Existing conflicting VIBs – remove old NSX-V VIBs before installing NSX-T. (4) Network connectivity issues – ESXi host cannot reach NSX Manager on port 443. (5) Disk space – insufficient space on ESXi host boot partition. Fix: resolve the root cause, then retry host preparation from NSX Manager: System > Fabric > Nodes > Host Transport Nodes > [host] > Actions > Resolve.

Q 258

What is NSX-T Syslog configuration?

Answer

Configure NSX-T syslog forwarding to a central log server: NSX Manager syslog: System > Settings > Syslog Servers > Add (specify server IP, port, protocol UDP/TCP/TLS, log level). Edge node syslog: set syslog from Edge CLI: set logging-server proto udp level info. ESXi host syslog (for DFW logs): configure in vSphere via NSX Manager automatically (host preparation includes syslog for NSX VIB logs). DFW flow logs: configure in Security > Distributed Firewall > per-rule logging. Ensure syslog server has capacity for high-volume DFW logs. Use structured logging (syslog with JSON format where supported) for easier SIEM parsing.

Q 259

What are NSX-T performance monitoring metrics?

Answer

NSX-T performance metrics to monitor: Manager Cluster: CPU usage (< 80%), memory usage (< 90%), disk usage of /var/log (< 80% – rotate logs if high), API response time. Edge Nodes: CPU per core (DPDK cores should be < 80%), data plane throughput (monitor in Edge CLI: get dataplane stats), BGP session stability (flapping indicates network issues), VPN tunnel statistics (bytes/sec, dropped packets). Transport Nodes: TEP packet drops (indicates MTU issues), DFW rule evaluation latency, ESXi host CPU (DFW adds ~5-10% CPU overhead per VM vNIC). Use NSX Manager > System > Overview for high-level health, and Aria Operations NSX MP for detailed historical metrics.

Q 260

What is NSX-T packet capture on Edge?

Answer

Packet capture on NSX-T Edge nodes is used for deep troubleshooting of traffic flowing through Edge (BGP, VPN, NAT, LB). Steps: SSH to Edge node. Identify the interface: get interfaces (shows management eth0, fp-eth0/1/2). Enable capture: debug packet capture interface expression 'host 10.0.0.1' (tcpdump-like syntax). Captures are limited in size – export to NSX Manager or SCP off. Alternative: NSX Manager Central CLI has a limited capture capability. For data plane interfaces (fp-eth), use: debug dataplane packet-capture. Note: packet capture on production Edge nodes generates overhead – use it briefly and only for active troubleshooting. Disable immediately after capturing required packets.

Q 261

What is the NSX-T configuration import/export?

Answer

NSX-T supports exporting and importing configuration via the API for backup and migration purposes. Export: GET /policy/api/v1/infra (exports the entire Policy infrastructure configuration as JSON). Import: PUT/PATCH /policy/api/v1/infra (imports configuration from JSON). This is useful for: migrating NSX-T configuration between environments (dev to prod), version-controlled configuration management (GitOps for NSX-T config), and disaster recovery (restore NSX-T configuration to a new deployment). The export does not include system-level settings (certificates, users) – those are handled separately. Use the NSX-T backup mechanism for full system recovery.

Q 262

What is NSX-T System Health Indicators?

Answer

NSX Manager System Health indicators (Home page): Manager Nodes (green = all 3 managers healthy, cluster VIP active), Transport Nodes (green = all host/edge nodes stable, no pending config), Edge Cluster (green = all Edge nodes up and BGP established), Alarms (count of Critical/High/Medium alarms — should be zero in healthy environment). For detailed health: System > System Overview shows each node, service status (NSX Manager services: corfu, rabbitmq, search, proxy), disk and memory utilization. Health check scripts: run `get manager-health` on NSX Manager CLI (SSH to manager) for a quick all-component health summary. Automate health checks via API: `GET /api/v1/cluster/nodes`.

Q 263

What is the difference between T0 and T1 failover modes?

Answer

T0 Failover Mode (affects entire T0 gateway): Active-Standby (one Edge active, one standby — failover triggers on BFD or link failure), Active-Active with ECMP (all Edges active simultaneously, physical ECMP distributes traffic). T1 Failover Mode (affects individual T1 SR instances): Active-Standby (T1 SR on one Edge is active, SR on another Edge is standby — each T1 can be on a different active/standby Edge for distribution). In practice: configure all T1s to use the same Edge cluster as the T0, and NSX-T automatically distributes T1 SR instances across Edge nodes for load balancing within the cluster.

Q 264

What is NSX-T VPN and NAT interaction?

Answer

When NSX-T VPN (IPsec) coexists with NAT on the same T0/T1 gateway, traffic order matters: For outbound VPN traffic (overlay VMs sending to VPN-connected remote networks): NAT is typically bypassed for traffic destined for VPN remote subnets using a No-SNAT rule (explicitly matching VPN remote subnets and setting action to 'No SNAT'). For inbound VPN traffic (from remote sites to internal overlay VMs): DNAT may be needed if remote sites must reach VMs using public/translated IPs. Configure No-SNAT rules before SNAT rules in the NAT rule priority order to ensure VPN traffic is not accidentally translated. Always test VPN traffic flow after configuring NAT changes.

Q 265

What is NSX-T Port Mirroring (ERSPAN)?

Answer

NSX-T supports Port Mirroring via ERSPAN (Encapsulated Remote Switched Port Analyzer) on the N-VDS/VDS. Configure mirror sessions in NSX Manager to copy traffic from specific segments or logical ports to a remote ERSPAN destination (a monitoring VM or physical appliance). Use cases: IDS/IPS that requires full packet capture, network forensics, application performance monitoring, security analysis. ERSPAN encapsulates mirrored packets in GRE and sends them to the configured collector IP. ERSPAN adds overhead — ensure the uplinks have enough bandwidth. Configure via NSX Manager: Networking > Segments > [Segment] > Port Mirroring Sessions, or via vSphere VDS Port Mirroring configuration.

Q 266

What is NSX-T forwarding table verification?

Answer

Verify NSX-T forwarding tables to troubleshoot routing/connectivity: On Edge node (for T0/T1 SR routing): SSH to Edge, enter T1/T0 VRF context, then get forwarding-table (shows ARP, routing forwarding entries). On ESXi host (for DR distributed routing): net-vdr --lif list -V (lists logical interfaces on the host's DR), net-vdr --route list -V (shows routing table on host DR). For overlay MAC/IP tables on ESXi host: net-dvs -l (lists N-VDS ports and MACs), nsxcli ens realization (shows segment realization on host). These commands help verify that control plane routing information has been correctly programmed into the data plane on each transport node.

Q 267

What is NSX-T manager memory pressure handling?

Answer

NSX Manager memory pressure handling: if NSX Manager VM runs low on memory (above 85-90% usage), performance degrades — API calls slow down, UI becomes unresponsive, and the manager may fail to push configuration to transport nodes. Causes: undersized NSX Manager (Small in production), memory leak in NSX services (fixed in patches), or excessive audit log accumulation. Resolution: (1) Immediate: SSH to NSX Manager, run service manager restart (restarts NSX services, releases memory — brief management plane interruption, data plane continues). (2) Short-term: clear old audit logs, purge old NSX Manager backups. (3) Long-term: resize NSX Manager VM to Large appliance size, apply latest NSX patches. Always monitor NSX Manager memory via NSX Manager UI or Aria Operations.

Q 268

What is NSX-T Global Search functionality?

Answer

NSX Manager Global Search allows searching across all NSX-T objects — segments, VMs, ports, rules, groups, gateways, IP addresses — from a single search bar. Enter any keyword (VM name, IP address, MAC address, rule ID, segment name) and NSX Manager searches across all object types and returns matching results with direct links. Global Search is invaluable for: quickly finding which segment a VM is on, identifying which DFW rules reference a specific group or IP, locating an IP address across the entire NSX environment, and navigating large NSX deployments with hundreds of objects. Access via the search icon in NSX Manager top navigation bar. API equivalent: GET /api/v1/search?query=.

Q 269

What is NSX-T Data Plane Monitor (DPM)?

Answer

NSX-T Data Plane Monitor (available via NSX Manager UI: Plan and Troubleshoot > Data Plane Monitor) provides real-time visibility into the NSX-T data plane health across all transport nodes: TEP connectivity matrix (which hosts can reach which other hosts via overlay), latency and packet loss statistics between TEPs, and Geneve tunnel health status. DPM continuously tests overlay connectivity using ICMP probes between all transport node TEPs and Edge node TEPs. Failures or high latency are highlighted in the DPM dashboard — critical for identifying: physical network issues affecting overlay (packet loss on TEP VLAN), MTU problems (large packet drops), and misconfigured TEP VLANs on specific hosts.

Q 270

What is NSX-T in-band management vs out-of-band management?

Answer

In-band management: NSX Manager is accessed via the same network that carries workload traffic. Risk: if network issue affects workload network, management access is also lost. Out-of-band management (recommended): NSX Manager management interface is on a dedicated management network (separate physical NICs, dedicated management VLAN) that is not shared with workload or overlay traffic. If the overlay network has issues, management access via the dedicated management network is unaffected — administrators can still troubleshoot. Similarly, NSX Edge management interface (eth0) should be on the dedicated management network, not the data plane VLANs. Always design dedicated management networks for all infrastructure components (NSX Manager, vCenter, ESXi management) following out-of-band management principles.

NSX-T Multi-Cloud & Advanced Topics

30 Q&A;

Q 271

What is VMware NSX-T on VMware Cloud on AWS (VMC)?

Answer

VMware Cloud on AWS (VMC) runs the full VMware SDDC stack (vSphere + vSAN + NSX-T) on dedicated AWS bare-metal hardware managed by VMware. NSX-T in VMC provides: overlay networking for VMs in the SDDC, T0/T1 gateways for routing, DFW for security, native integration with AWS services via AWS ENI (Elastic Network Interface) for low-latency access to RDS, S3, Lambda. NSX-T configuration in VMC is done via the SDDC console (not directly via NSX Manager CLI). HCX (Hybrid Cloud Extension) uses NSX-T L2 VPN for workload migration between on-premises and VMC.

Q 272

What is VMware NSX-T on Azure VMware Solution (AVS)?

Answer

Azure VMware Solution (AVS) runs VMware SDDC (vSphere + vSAN + NSX-T) on dedicated Azure bare-metal servers, jointly managed by VMware and Microsoft. NSX-T in AVS provides overlay networking, T0/T1 gateways, and DFW for VMs in the AVS SDDC. AVS NSX-T integrates with Azure native services via Azure ExpressRoute and Azure Virtual Network Gateway. Administrators manage NSX-T configuration via the AVS NSX Manager. Consistent NSX-T policies can be extended from on-premises to AVS using HCX and NSX Federation, enabling true hybrid cloud networking.

Q 273

What is VMware HCX and how does it use NSX-T?

Answer

VMware HCX (Hybrid Cloud Extension) is a workload mobility and migration platform that uses NSX-T technologies: HCX Network Extension uses NSX-T L2 VPN technology to stretch Layer 2 segments between on-premises and cloud (VMC/AVS), allowing VMs to maintain their IP addresses post-migration. HCX vMotion uses NSX-T overlay networking for live VM migration over WAN. HCX Bulk Migration uses disk replication over HCX-IX (Interconnect) appliances. HCX is the recommended tool for migrating VM workloads from on-premises to VMware Cloud environments while maintaining consistent NSX-T security policies.

Q 274

What is NSX-T Project (Multi-tenancy in NSX 4.x)?

Answer

NSX Projects (introduced in NSX 4.x) provide native multi-tenancy within a single NSX Manager cluster. Each project is an isolated namespace with its own networking resources (segments, T1 gateways), security policies (DFW rules), load balancers, and VPN configurations. Project administrators can only see and manage their own project resources. Projects use shared T0 gateways (managed by the infrastructure admin) for North-South connectivity. This enables multi-tenant NSX environments (e.g., multiple business units or customers sharing NSX infrastructure) without deploying separate NSX Manager clusters per tenant.

Q 275

What is NSX-T Cloud Service Manager (CSM)?

Answer

NSX Cloud Service Manager (NSX Cloud) extends NSX-T security policies to workloads running in public clouds (AWS, Azure) without NSX-T infrastructure in the cloud. NSX Cloud deploys lightweight NSX Agents on public cloud VMs that connect back to the on-premises NSX Manager. This enables: consistent DFW micro-segmentation policies to apply to public cloud VMs using the same NSX Manager UI/API, traffic visibility for cloud workloads in NSX Intelligence, and unified security policy management across on-premises and cloud without requiring VPN or VXLAN overlay to the cloud.

Q 276

What is NSX-T Data Center with Kubernetes (Tanzu)?

Answer

vSphere with Tanzu (Supervisor Cluster) uses NSX-T as the networking provider for both VM and container workloads: NSX-T provides the overlay network for Kubernetes pods (via NCP or Antrea-NSX integration), load balancer services for Kubernetes (NCP creates NSX-T VIPs automatically), network policy enforcement (Kubernetes NetworkPolicy translated to NSX-T DFW rules), and persistent storage networking. The Kubernetes namespace maps to NSX-T logical objects (segments, firewall policies). This provides a unified networking and security platform for both traditional VMs and modern containerized applications on the same vSphere/NSX-T infrastructure.

Q 277

What is NSX-T Compliance and Certification?

Answer

NSX-T has achieved multiple compliance certifications: FIPS 140-2 (Level 1) – cryptographic modules used in NSX-T (IPsec VPN, TLS) are FIPS-certified. Common Criteria EAL2+ – NSX-T DFW is evaluated under Common Criteria for security assurance. PCI-DSS – NSX-T micro-segmentation is widely used in PCI cardholder data environment (CDE) segmentation. HIPAA – NSX-T helps achieve HIPAA network security controls. SOC 2 – VMware's NSX-T cloud services have SOC 2 Type II certification. GDPR – NSX-T encryption and access controls support GDPR data protection requirements. The specific certifications vary by NSX-T version – check VMware's compliance documentation.

Q 278

What is the NSX-T capacity planning process?

Answer

NSX-T capacity planning steps: (1) Identify workload count (number of VMs/containers/bare-metal nodes). (2) Determine throughput requirements (East-West bandwidth per host, North-South bandwidth through Edge). (3) Size NSX Manager cluster (Small/Medium/Large based on transport node count). (4) Size Edge nodes (Small/Medium/Large/XL based on routing throughput, LB connections, VPN tunnels). (5) Plan Edge cluster redundancy (2+ nodes per cluster, N+1 or active-active). (6) Calculate TEP IP pool size (at least one IP per transport node, plus spares). (7) Plan firewall rule capacity (NSX-T supports up to 100K DFW rules per manager cluster). (8) Reference VMware Configuration Maximums guide for exact limits.

Q 279

What is NSX-T integration with vRealize Automation (Aria Automation)?

Answer

NSX-T integrates with VMware Aria Automation (vRA) via the NSX-T endpoint configured in Aria Cloud Assembly. When a cloud template deploys a VM, Aria Automation can automatically: create NSX-T segments, assign VMs to segments, apply security group tags (which trigger DFW rule inclusion), provision load balancer VIPs, and configure NAT rules. This enables Infrastructure-as-Code (IaC) for NSX-T network/security provisioning – network resources are created and destroyed alongside application VMs in automated deployments. The NSX-T Terraform provider provides the same capability for GitOps/CI-CD pipelines.

Q 280

What is NSX-T integration with Splunk?

Answer

NSX-T integrates with Splunk for security monitoring and SIEM via: Syslog forwarding (NSX Manager and Edge nodes forward security events, DFW flow logs, audit logs to Splunk via UDP/TCP syslog), VMware NSX-T Add-on for Splunk (pre-built Splunk app with dashboards for DFW events, IDS alerts, VPN status, BGP events), NSX Intelligence flow data export to Splunk for network behavior analytics, and Splunk SOAR integration (automated response to NSX-T IDS alerts — e.g., automatically tag a compromised VM and apply quarantine DFW policy via NSX-T API). This provides unified security visibility across the entire data center.

Q 281

What is NSX-T with SD-WAN?

Answer

VMware SD-WAN (VeloCloud) integrates with NSX-T for branch office connectivity. In SD-WAN + NSX-T designs: Branch offices have VeloCloud SD-WAN Edge devices that connect to SD-WAN Gateways in the data center. NSX-T T0 gateways peer with SD-WAN gateways via BGP to exchange routes between the NSX overlay and SD-WAN overlay. Security policies (NSX-T DFW) can be extended to branch workloads via SD-WAN. VeloCloud provides application-aware WAN optimization and traffic steering; NSX-T provides data center micro-segmentation and security. Together they provide end-to-end consistent networking from branch to data center.

Q 282

What is NSX-T Data Flow Monitoring (DFW Logging)?

Answer

NSX-T DFW supports per-rule logging — when enabled, each connection that matches a DFW rule generates a log entry containing: source IP/MAC, destination IP/MAC, protocol, port, action (Allow/Drop), rule ID, and VM name. DFW logs are sent to the ESXi host's syslog and should be forwarded to a central syslog/SIEM. Enabling DFW logging on all rules generates significant log volume — log selectively on critical rules (default drop, inter-zone rules). DFW log format can be parsed by Splunk, Elastic, and other SIEMs using the NSX-T Splunk Add-on or custom parsing. Flow logs are essential for security incident investigation.

Q 283

What is NSX-T Greenfield vs Brownfield deployment?

Answer

Greenfield NSX-T deployment: deploying NSX-T in a new vSphere environment with no existing workloads. Simpler — design the network architecture from scratch using NSX-T best practices (Policy API, overlay segments, T0/T1 hierarchy, DFW). Brownfield NSX-T deployment: adding NSX-T to an existing vSphere environment with running VMs on traditional VLANs. Challenges: existing VMs must be migrated from VLAN port groups to NSX-T overlay segments (or VLAN-backed NSX-T segments as a transitional step), NSX-T N-VDS must coexist with existing vDS, security policies must be gradually applied without disrupting production. VMware provides migration tools and documented processes for brownfield deployments.

Q 284

What are the NSX-T licensing tiers?

Answer

NSX-T (now VMware NSX) licensing tiers: NSX Standard: DFW micro-segmentation, L2/L3 overlay, T0/T1 routing, basic LB, IPsec VPN. NSX Professional: adds advanced DFW features (Context Profiles, App-ID), Gateway Firewall. NSX Advanced: adds Distributed IDS/IPS, URL Filtering, advanced LB features, NSX Intelligence. NSX Enterprise Plus: adds NSX Federation, advanced multi-site features, all Advanced features. NSX Advanced Threat Prevention: adds AI/ML-based NDR, Malware Prevention, DNS Security (sold as add-on). Under Broadcom VCF/VVF bundles, NSX is included and specific tier depends on the bundle level. Always verify current licensing with VMware/Broadcom licensing team.

Q 285

What is NSX-T on Google Cloud VMware Engine (GCVE)?

Answer

Google Cloud VMware Engine (GCVE) runs VMware SDDC (vSphere + vSAN + NSX-T) on Google Cloud dedicated bare-metal servers. NSX-T in GCVE provides overlay networking, T0/T1 gateways for routing, DFW for micro-segmentation, and VPN connectivity. GCVE NSX-T integrates with Google Cloud services via Cloud Interconnect and VPC peering for accessing GCP-native services (Cloud SQL, BigQuery, GCS). NSX-T configuration in GCVE is managed through the GCVE portal or directly via NSX Manager. Consistent NSX-T policies between on-premises and GCVE are achieved through NSX Federation or HCX-based connectivity.

Q 286

What is NSX-T Automation with Terraform?

Answer

The VMware NSX-T Terraform Provider (terraform-provider-nsxt, available at registry.terraform.io) allows declarative Infrastructure as Code management of NSX-T resources. Key resources: `nsxt_policy_segment` (create overlay/VLAN segments), `nsxt_policy_tier0_gateway` / `nsxt_policy_tier1_gateway` (T0/T1 gateways), `nsxt_policy_security_policy` (DFW policies and rules), `nsxt_policy_group` (security groups with dynamic membership), `nsxt_policy_nat_rule` (NAT rules), `nsxt_policy_lb_virtual_server` (LB virtual servers). Terraform enables GitOps for NSX-T – all network/security changes are code-reviewed, version-controlled, and applied consistently across environments.

Q 287

What is NSX-T integration with vRealize Operations?

Answer

VMware Aria Operations (vRealize Operations) NSX-T Management Pack provides: real-time monitoring of NSX-T components (Manager cluster, Edge nodes, transport nodes), performance dashboards (throughput, packet drops, BGP session stability, DFW rule evaluation time), capacity planning for NSX-T (forecast when resource limits will be reached), anomaly detection (alert when NSX-T metrics deviate from baseline), and troubleshooting workflows (correlated alerts across vSphere and NSX-T). Install the NSX-T Management Pack in Aria Operations > Marketplace. Connect NSX Manager as an adapter instance. Provides the operational visibility needed for large-scale NSX-T environments.

Q 288

What is NSX-T content pack for vRealize Log Insight?

Answer

VMware Aria Log Insight (vRealize Log Insight) NSX-T Content Pack provides: pre-built dashboards for NSX-T log analysis (DFW events, BGP changes, Edge node events, VPN tunnel status), custom alerts for critical NSX-T events (Edge node disconnected, BGP session down, DFW rule change), and structured log parsing for NSX-T syslog format. Install from Log Insight Content Pack marketplace. Configure NSX Manager and Edge nodes to forward syslog to the Log Insight server (UDP/TCP 514 or 9000 for LI agent). For high-volume DFW log environments, deploy Log Insight at scale (cluster mode) to handle the log throughput. Log Insight is the primary tool for NSX-T log analysis and incident investigation.

Q 289

What is NSX-T REST API authentication?

Answer

NSX-T REST API authentication methods: (1) Basic Authentication: send Base64-encoded username:password in the Authorization header with every API call. Simple but not recommended for production (password in every call). (2) Session-based Authentication: POST to /api/v1/sessions to create a session (returns session cookie), then include the cookie in subsequent calls (valid for 24 hours by default). More secure than basic auth. (3) Principal Identity (Certificate-based): create a PI in NSX Manager with a client certificate; authenticate API calls by signing requests with the certificate. Used by Terraform, vRA, Kubernetes NCP for machine-to-machine authentication. No password required — most secure for automation. (4) vIDM Integration: authenticate via VMware Identity Manager (SAML/OAuth).

Q 290

What are NSX-T design best practices?

Answer

NSX-T design best practices: Deploy 3-node NSX Manager cluster on separate hosts with anti-affinity DRS rules. Use Policy API (not Manager API) for all new configurations — simpler, future-proof. Separate TEP, management, vMotion, and vSAN traffic onto dedicated VLANs and VMkernel ports. Use jumbo frames (MTU 9000) end-to-end for overlay traffic. Deploy 2+ Edge nodes per cluster; use active-active T0 with ECMP for maximum throughput. Create T1 gateway per application or tenant for routing isolation. Use tag-based DFW groups for dynamic, automated security policy. Enable DFW logging on critical rules only (default drop, inter-zone). Implement NSX Intelligence before enabling Default Drop policy. Regular backups, patching, and health checks are mandatory.

Q 291

What is NSX-T support for IPv6?

Answer

NSX-T supports IPv6 in overlay networking: IPv6 segments (dual-stack or IPv6-only), T0/T1 gateways with IPv6 routing (static, BGP, OSPFv3), IPv6 distributed routing (DR component handles IPv6 routing on each host), IPv6 DFW rules (source/destination groups with IPv6 addresses and CIDRs), NAT64 (translate IPv6 workloads to IPv4 for compatibility with legacy systems). IPv6 support in NSX-T has matured significantly from version 3.0 onwards. Not all features have full IPv6 parity with IPv4 — check the specific NSX-T version's release notes for IPv6 feature support matrix. The management plane (NSX Manager) can be configured with IPv4, IPv6, or dual-stack management IPs.

Q 292

What is NSX-T Export/Import of Firewall Rules?

Answer

NSX-T allows exporting and importing DFW rules for backup, audit, and migration: Export: NSX Manager > Security > Distributed Firewall > Export (downloads all DFW policies as a CSV or JSON file). Import: Import allows adding rules from a previously exported file – useful for restoring rules after accidental deletion or migrating rules between NSX-T environments. Rules export includes: rule name, source/destination groups, services, action, and metadata. Groups themselves are not exported – they must exist in the target environment before importing rules that reference them. For full configuration backup, use the NSX-T file-based backup (SFTP) rather than rule export.

Q 293

What is NSX-T and CrowdStrike integration?

Answer

NSX-T integrates with CrowdStrike Falcon via the NSX-T Service Insertion framework. CrowdStrike can register as a partner service in NSX-T. DFW rules can redirect traffic to CrowdStrike inspection engines for: EDR (Endpoint Detection and Response) telemetry, threat intelligence correlation (compare VM network traffic with CrowdStrike threat indicators), and automated response – when CrowdStrike detects a compromise, it can call the NSX-T API to apply a quarantine tag to the affected VM, immediately isolating it via DFW. This creates a closed-loop automated security response between CrowdStrike detection and NSX-T enforcement.

Q 294

What is NSX-T VMware Cloud on Dell (VCDD)?

Answer

VMware Cloud on Dell (formerly Project Dimension) brings the VMware Cloud experience to on-premises Dell hardware. Customers receive a fully managed VMware SDDC (vSphere + vSAN + NSX-T) on Dell hardware in their own data center, managed as a service by VMware/Dell. NSX-T in VCDD provides the same networking and security capabilities as on-premises NSX-T, with the operational simplicity of a managed service. Configuration is done via the SDDC management console. NSX-T policies created in VCDD can be federated with other NSX-T environments (on-premises or cloud) for consistent policy management.

Q 295

What is NSX-T Workspace ONE integration?

Answer

NSX-T integrates with VMware Workspace ONE (formerly AirWatch) for User and Device Identity-based security: Workspace ONE UEM provides device identity and compliance status. NSX-T Identity Firewall (IDFW) can use Workspace ONE device compliance data – for example, only allow managed/compliant devices to access sensitive VMs. A non-compliant device (missing patches, AV not running) would be denied access even if the user has valid credentials. This integration requires VMware Identity Manager (vIDM) as the federation layer. The result is a complete Zero-Trust framework where both user identity (from AD) and device compliance (from Workspace ONE) are required for network access.

Q 296

What is NSX-T and Terraform state management?

Answer

When using Terraform with the NSX-T provider, Terraform maintains a state file that maps Terraform resource definitions to actual NSX-T objects (by their IDs). Key considerations: store the Terraform state file remotely (S3, Terraform Cloud, GitLab) – not locally – so multiple team members can collaborate and the state is not lost if a local machine fails. Use Terraform workspaces for environment isolation (dev, staging, production). Implement Terraform CI/CD: code review via pull requests, automated plan (terraform plan shows proposed changes), apply only after approval. If NSX-T objects are modified outside Terraform (via UI), run terraform refresh to sync state. Never manually delete NSX-T objects managed by Terraform – always use terraform destroy.

Q 297

What is VMware Aria Automation integration with NSX-T?

Answer

VMware Aria Automation (vRA) integrates with NSX-T via a configured NSX-T endpoint in Aria Cloud Assembly. Cloud Templates (Blueprints) define application topology including: VM compute resources, NSX-T network profiles (auto-creates segments on demand), NSX-T security groups (auto-applies tags to deployed VMs for DFW rule inclusion), LB virtual servers (auto-provisioned for web-tier deployments), and NAT rules (auto-configured for public access). When a developer deploys an application via the Aria Automation Service Catalog: Aria orchestrates vCenter VM creation AND NSX-T network/security provisioning atomically – the entire stack is deployed as a unit. Deprovisioning removes both VMs and NSX-T resources.

Q 298

What is NSX-T upgrade path and version compatibility?

Answer

NSX-T upgrade version compatibility rules: you cannot skip major versions – must upgrade sequentially (e.g., 3.1 > 3.2 > 3.3, not 3.1 > 3.3 directly). NSX Manager must be upgraded before Edge nodes, and Edge nodes before host transport nodes. NSX-T and vSphere version compatibility is defined in the VMware Product Interoperability Matrix (vmware.com/resources/compatibility/sim). NSX-T 3.x is supported on vSphere 6.7 U3+ and vSphere 7.x. NSX 4.x requires vSphere 7.0 U3+ or vSphere 8.x. Always check interoperability matrix before planning upgrades. Keep all components (Manager, Edge, Hosts) within 1 major version of each other – running mismatched versions long-term is unsupported.

Q 299

What is NSX-T Segment spanning vs local segments?

Answer

NSX-T Segments can span the entire Transport Zone (available on all hosts in the transport zone) or be limited to specific hosts. Spanning types: Full Transport Zone Span (default – segment is available on all hosts in the transport zone, maximum flexibility for VM placement), Partially Span (admin-defined – segment only available on specific ESXi clusters, useful for restricting VM placement for compliance or performance). For most deployments, full transport zone spanning is used. Limiting segment span is used in: multi-tenant environments (tenant A's segments should not exist on tenant B's hosts), compliance requirements (PCI VMs must only run on certified hosts), and performance optimization (keeping high-I/O VMs on hosts with faster storage).

Q 300

What is NSX-T end-to-end automation workflow?

Answer

A complete NSX-T automation workflow: (1) Source control (Git): store all NSX-T configuration as Terraform HCL or Ansible YAML. (2) CI/CD pipeline (GitLab/GitHub Actions): on pull request, run 'terraform plan' – show proposed NSX-T changes without applying. Automated tests validate plan (no default Allow rules, required tags present). (3) Code review: team reviews proposed changes (4-eyes principle for production). (4) Merge and apply: on merge to main, pipeline runs 'terraform apply' – creates/updates NSX-T objects. (5) Post-apply verification: automated tests verify realization state (all objects stable), connectivity tests (Traceflow API calls), and security tests (verify blocked flows stay blocked). (6) Monitoring: Aria Operations alerts on unexpected deviations from desired state. This GitOps workflow eliminates manual NSX-T changes and provides full audit trail.

This guide covers 300 VMware NSX-T interview questions across 10 core topic areas — Architecture, Switching, Routing, Edge Nodes, DFW, Load Balancer, VPN, Security, Operations, and Multi-Cloud. Combine with hands-on NSX-T lab practice for best results.

Created by Akhilesh Chaudhari

akhileshchaudhari.com | [TechShiksha](#) | 2026 Edition